

UNITEC INSTITUTE OF TECHNOLOGY

MASTER THESIS

**Designing a Data-Driven Congestion Pricing System
for Auckland Using AI**

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Abstract

The Auckland road network experiences significant congestion pressures that affect commuters, freight operators, and the wider city economy. This thesis designs and evaluates a proof-of-concept data-driven congestion pricing system for Auckland that integrates traffic simulation, short-term forecasting, adaptive toll pricing, behavioural demand modelling, congestion feedback, and MQTT-based publish-subscribe communication. Traffic data were generated using the Simulation of Urban Mobility (SUMO) platform with an OpenStreetMap-based Auckland road network. The 48-hour simulation produced edge-level statistics at 60-second resolution, which were cleaned, aggregated into 10-minute observations, and summarised as a corridor-level traffic-state indicator containing 287 observations. The 287 labelled observations were chronologically divided into 229 training observations (80%) and 58 held-out test observations (20%). ARIMA, XGBoost, and Long Short-Term Memory (LSTM) models were evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) on the common held-out test set. On the common 58-observation held-out test set, ARIMA achieved the lowest overall forecasting error, with MAE of 0.1954 seconds and RMSE of 0.2508 seconds, followed closely by LSTM with MAE of 0.1962 seconds and RMSE of 0.2602 seconds. XGBoost recorded MAE of 0.2494 seconds and RMSE of 0.3235 seconds, while a naive persistence baseline achieved MAE of 0.2453 seconds and RMSE of 0.3425 seconds. A free-flow reference of 1.3140 seconds was estimated from the 5th percentile of the training observations only. Using the 58 held-out ARIMA forecasts within the dynamic-pricing scenario, all 58 test intervals exceeded the configured toll-activation threshold, producing an average applied toll of NZD 2.0886. Under the assumed elasticity and feedback parameters, the scenario model estimated an average demand reduction of 4.1772% and an average traffic-state reduction of 3.3418%. These values represent proof-of-concept scenario outputs rather than observed Auckland traffic or behavioural effects. The prototype was extended with an MQTT-based IoT communication layer to demonstrate publish-subscribe data streaming and a Gradio dashboard for visualising datasets, congestion calculations, dynamic tolls, and scenario comparisons. The work demonstrates that simulation, forecasting, pricing, demand response, congestion feedback, and publish-subscribe communication can be combined within a coherent and reproducible Intelligent Transport System prototype. It should not be interpreted as a policy recommendation and instead provides a foundation for future research using real Auckland traffic data, calibrated route-level travel times, improved behavioural models, and equity analysis.

Keywords: congestion pricing, Auckland, intelligent transportation systems, travel-time forecasting, SUMO, ARIMA, XGBoost, LSTM, dynamic tolling, IoT, MQTT, demand response

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
AMQP	Advanced Message Queuing Protocol
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
AT	Auckland Transport
CoAP	Constrained Application Protocol
CSV	Comma-Separated Values
TCN	Temporal Convolutional Network
GNN	Graph Neural Network
GIS	Geographic Information System
GPS	Global Positioning System
IoT	Internet of Things
ITS	Intelligent Transportation Systems
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MQTT	Message Queuing Telemetry Transport
NZTA	New Zealand Transport Agency
OSM	OpenStreetMap
RMSE	Root Mean Square Error
SUMO	Simulation of Urban Mobility
UI	User Interface
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Background

There are many big cities like Auckland [1], [2] which are suffering from congestion that is real and tangible. Population growth and urbanization together with increased travel of vehicles have resulted in increased number of vehicles on roads. When there are more travelers than what the road system is capable of serving, the citizens face delays and uncertainty about travel times as well as increased fuel consumption and emissions.

There are other methods for congestion management besides the traditional road widening and toll/paced/retimed traffic signals, and they can be effective in some situations [18], [19]. But such methods are ineffective because they do not adapt to the constantly changing traffic conditions throughout the day. The static toll can be too expensive when the volume of traffic is low, and too low when traffic is high. That's why dynamic congestion pricing is an area of interest in transport [10]–[12]. The intention is to change the price to make the price signal (the price of using the road) more representative of the traffic situation.

In this thesis, the term artificial intelligence is used in the broad sense adopted by intelligent transportation systems research, encompassing statistical forecasting, machine learning, and deep learning methods evaluated within a data-driven decision pipeline [81][86]. The AI contribution of this work lies not only in the individual forecasting models tested, but in the integrated system architecture that connects simulation, prediction, pricing, behavioural modelling, and real-time communication. This reflects the definition of intelligent transport systems as systems that apply information processing, communication, and decision-support technologies to transport management a definition that encompasses the full prototype developed in this thesis[81] [84].

The problem becomes more challenging when using machine learning and traffic simulation because it enables look-ahead rather than just look-back. A traffic simulation can be set up using SUMO and short-term travel times can be forecasted. Also, IoT and lightweight protocols such as MQTT demonstrate how traffic data could be communicated across sensors, systems and decision support

models in this real-time system [81]–[84], [93], [94].

The contribution of this thesis is to provide a single integrated prototype that ties these pieces together for an Auckland-based system. So, this research aims to build an integrated workflow between simulation, data processing, forecasting, pricing, demand response, congestion feedback and system demonstration.

1.2 Research Objectives

1.2.1 Main Aim

The purpose of this study is to develop, prototype and test a data-driven dynamic congestion pricing system for Auckland [1], [2], [87]–[90]. This research integrates traffic simulation, prediction, pricing and a rudimentary real-time framework. The objective is not to develop the final congestion pricing policy, but to see whether a viable technical workflow is available to help implement an adaptive and predictive congestion management policy in New Zealand transport.

1.2.2 Specific Objectives

To achieve the main aim, the following specific objectives are defined:

1 Model Development and Evaluation

- Training and evaluating multiple traffic forecasting models (such as ARIMA, XGBoost and LSTM) with traffic simulation data.
- To assess the performance of the model using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).
- The aim of the process is to choose the optimal model for short-term traffic forecasting for congestion pricing.

2. Data Processing and System Pipeline Design

- To simulate traffic in SUMO for 48 hours.
- To obtain traffic statistics at edge level at 60-second resolution.
- To generate 10-minute travel time for forecasting at a corridor level.
- To organise data into raw, cleaned, labelled, training, validation and testing data sets.

3. Dynamic Pricing Model Development

- To create a congestion index from the predicted aggregated edge-based corridor travel-time indicator compared to the simulated free-flow baseline of the aggregated indicator.
- To calculate the simulated free-flow baseline of the aggregated indicator as the 5th percentile of the indicator values.
- To create a pricing function that will allow the implementation of congestion based variable tolls.
- To take into account policy factors such as toll cap for realism and equity.

4. Demand Response and Feedback Analysis

- To use elasticity assumptions to model driver behaviour.
- To find out the effect of dynamic pricing on demand.
- To assess the effects of decreased demand on congestion.
- To propose techniques to measure travel time and system efficiency benefits.

5. System Integration and Real-Time Extension

- To develop an IoT communication model using data streaming and MQTT broker.
- To publish and subscribe data structure.
- To use Gradio to make a user interface (UI) to show the system.
- To consider the prototype’s potential to be scaled up to a practical application.

1.3 Questions & Challenges

This study is guided by the following research questions, which are aligned with the proof-of-concept scope of the implemented system: [1]–[20], [40]–[61], [73]–[86]:

1. How accurately can ARIMA, XGBoost, and LSTM forecast the simulation-derived aggregated edge-based corridor traffic-state indicator 10 minutes ahead on a held-out chronological test period?
2. How can the forecast traffic-state indicator be translated into a configurable dynamic congestion-pricing scenario using a simulated free-flow baseline, congestion-index thresholds, and a bounded toll function?

3. Under the stated demand-elasticity and congestion-feedback assumptions, what modelled changes in travel demand and the simulated congestion indicator are produced by the dynamic pricing scenarios?
4. How can SUMO-based traffic simulation, forecasting models, dynamic pricing, MQTT communication, and a Gradio dashboard be integrated into a reproducible end-to-end intelligent transport prototype?
5. What methodological, technical, behavioural, and policy limitations must be addressed before the prototype could be evaluated for possible real-world application in Auckland?

These questions deliberately focus on prototype implementation and model-derived scenario evaluation. The study does not attempt to establish causal effects on real Auckland traffic, observed driver behaviour, or operational congestion-pricing performance.

1.4 Scope of Study

This research is limited to the design and evaluation of a simulation-based proof-of-concept for dynamic congestion pricing using an Auckland-network-based SUMO environment [73]–[80], [102]. The study does not evaluate an operational Auckland congestion-pricing scheme and does not use calibrated real-world Auckland traffic observations as the primary modelling dataset.

The traffic variable used throughout the prototype is a simulation-derived aggregated edge-based corridor traffic-state indicator rather than a complete end-to-end corridor journey time [93], [94]. The indicator is used to compare forecasting models, calculate a simulated congestion index, and demonstrate how a configurable pricing scenario can be generated from held-out ARIMA forecasts. An earlier implementation used a one-step shifted traffic-state proxy, but the corrected pricing summary reported in Chapter 4 uses the common 58-observation held-out period.

The behavioural component is also limited to scenario modelling. Changes in travel demand are estimated using a stated elasticity assumption, and the resulting congestion change is calculated using an algebraic feedback model. These values therefore represent model-derived scenario estimates and should not be interpreted as observed driver behaviour, measured congestion reduction, or causal evidence of policy effectiveness.

The MQTT and Gradio components are implemented as proof-of-concept system-integration artefacts. MQTT demonstrates local publish-subscribe communication, while Gradio provides an interactive interface for visualising data, pricing parameters, and scenario outputs. The study does not evaluate production-level latency, reliability, cybersecurity, large-scale deployment, or live Auckland sensor integration.

Accordingly, the scope of the research is to demonstrate the technical integration, transparency,

and reproducibility of the proposed prototype and to identify the requirements for stronger future validation using real traffic data, larger simulations, calibrated behavioural models, and formal policy evaluation.

1.5 Significance of the Study

The main contribution of this research is the implementation and integration of established transport simulation, forecasting, pricing, behavioural modelling, communication, and visualisation methods within a single Auckland-oriented proof-of-concept system.

The study does not propose a new forecasting algorithm or a new congestion-pricing theory. Instead, its contribution lies in demonstrating how existing methods can be combined into a transparent and reproducible workflow that begins with SUMO-generated traffic data and continues through data processing, short-term forecasting, congestion-index calculation, configurable pricing scenarios, modelled demand response, algebraic congestion feedback, MQTT communication, and Gradio-based visualisation [81]–[84].

A second contribution is the comparative implementation of ARIMA, XGBoost, and LSTM under the same short-term forecasting objective. This provides evidence about how different model families behave when applied to a small, highly aggregated simulation-derived traffic-state dataset, while also highlighting the importance of model simplicity, interpretability, and data characteristics.

A third contribution is the development of an interactive prototype that allows users to inspect datasets and explore how pricing assumptions influence model-derived scenario outputs. This improves transparency because the relationship between the forecast traffic-state input, congestion index, toll assumptions, and estimated behavioural response can be inspected directly rather than remaining hidden within separate scripts.

The research is therefore significant primarily as an applied systems-integration study and as a foundation for future work. Any practical application to Auckland would require calibration against observed traffic data, improved behavioural modelling, network-level simulation, equity analysis, policy evaluation, and operational system testing [10]–[15].

1.6 Thesis Structure

This thesis is structured in the following way:

Chapter 1 provides the research background, objectives, research questions, and scope

Chapter 2 includes an extensive literature review on congestion pricing, traffic forecasting and intelligent transport systems.

Chapter 3 explains the data, data processing methods, and evaluation measures of the study.

Chapter 4 The proposed methodology, which includes simulation, forecasting models, pricing algorithm and system integration.

Chapter 5 will discuss the results, conclusions, limitations and future research directions.

Chapter 2

Literature Review

2.1 Introduction

One of the common problems in cities is the congestion of the city, however, the causes and solutions of this congestion are not the same in each city. The causes of some of the congestion are due to population growth and land use, while limited capacity, peak hour congestion, incidents and failure to identify suitable alternatives to some trips are the causes of the rest of the congestion. This has led to a new paradigm in congestion management, and today only the provision of new road space is viewed as a solution to congestion. Pricing, forecasting, simulation, and intelligent transport systems are also now being looked into by researchers and transport agencies. The major domains that are related to the current thesis are analyzed in detail in the current chapter, which is congestion pricing, traffic prediction, aggregating the data, IoT-Enabled transport systems, and demand response. This is intended to show how the current project can be incorporated into the existing project and the advantages of having an integrated Auckland prototype.

The review is organised to move from the broad transport problem to the specific technical components used in the prototype. First, congestion pricing is discussed as a policy and demand-management approach. Second, short-term traffic forecasting is reviewed because the pricing model depends on predicted rather than only observed traffic conditions. Third, data aggregation and corridor-level modelling are examined because the raw simulation outputs must be transformed into a stable modelling dataset. Fourth, intelligent transport systems and IoT communication are reviewed because a practical pricing system requires data exchange between sensors, models and user interfaces. Finally, behavioural response is discussed because a toll only reduces congestion if some travellers respond to the price signal.

This structure is important because the proposed project is not only a forecasting exercise [10]–[15], [40]–[61], [81]–[84]. It is an integrated system that connects simulation, prediction, pricing and communication. Therefore, the literature review evaluates both individual methods and the need for a complete workflow. This helps to identify why an Auckland-focused prototype can contribute to existing research even though each individual component has already been studied in other transport

contexts.

2.2 Congestion Pricing Approaches

Congestion pricing is a proven and established demand management strategy that can be used to cause economic incentives to alter the behaviour of drivers and other users of the road network during times of congestion [22]–[24]. Initial trials demonstrated within the prototype its practicality: Singapore’s Electronic Road Pricing system demonstrated that real-time data based tolling could lead to significant reduction in urban traffic congestion, and area-based charging systems were adopted in London, and Stockholm, and were shown to bring about measured decreases in traffic volumes and improvements in air quality [3]–[8]. Most of these systems, however, are based on fixed or static pricing models, which are not based on current traffic conditions. This can cause the scheme to be much less effective and less popular than intended because of either unexpectedly high traffic volumes during high-demand periods or unexpectedly low traffic volumes during low-demand periods, when fixed tolls would be far too high. This can result in the scheme being much less effective and acceptable to the public because traffic volumes are unexpectedly high during periods when the scheme is being used for the purpose it was designed to provide, and unexpectedly low during periods when traffic volumes are lower than required to justify upholding the scheme with fixed charges.

In recent years, a number of models have been suggested that would charge tolls dynamically depending on the level of congestion. These are usually based on traffic features like speed, density or travel time. Many current approaches are effective, but do not integrate with predictive models that help to predict future congestion.

2.2.1 Static and Dynamic Pricing

There are basically two types of congestion pricing: static and dynamic [10]–[17], [33]–[39]. Static pricing is setting the prices in advance for a certain period of time or in certain areas. This method is more simple to execute since the toll is foreseeable, and it could be clearly explained to the public. For instance, a city could have a first charge in the morning peak and a second charge during off-peak hours. The other advantage of static pricing is that the technology and rules that must be applied in order to determine the price are relatively straightforward for transport agencies.

Static pricing does have one important drawback, however: it is not directly responsive to the real time traffic. The congestion experienced can be quite different from day to day due to weather, incidents, road works, school holidays, public events and changes in travel behaviour. Therefore, a fixed toll might be too low if the road network is congested and too high if it is not. This mis-match may not only lower the effectiveness of the scheme but also diminish its acceptability to the public.

However, dynamic pricing is an attempt to overcome this with the ability to vary the toll based on traffic. For a dynamic system, the price may go up if there is expected congestion, and it may go down if there is little congestion. This benefits the price signal because the cost of congestion is more comparable to the actual congestion cost of the additional vehicle. Dynamic pricing is thus a more appropriate approach to intelligent transportation systems that utilize sensors, traffic models, and real-time data processing.

The difficulty is that dynamic pricing needs to have a greater level of technical understanding. The system should be capable of gathering/accepting traffic data, processing of this data in real time and with speed, congestion estimation, calculation of the toll, and communication with the user/operator. If any one step in this process is questionable, it is possible that the pricing decision will become uncertain or hard to defend. As a consequence, dynamic pricing is a policy issue as well as a data, modelling and system-integration issue.

2.2.2 International Congestion Pricing Examples

Lessons can be derived from case studies across the world with relevance to Auckland [3]–[8]. The most prominent among such cases is Singapore, where a changeover was made from manual area licensing scheme to electronic road pricing. In Singapore, its system of electronic road pricing proves technical feasibility by means of electronic vehicle identification, defined pricing rules, and constant calibration of policies based on real time traffic observations. Road pricing has proven to be more effective when applied within an overall transport policy rather than merely for raising revenues.

London’s congestion charging scheme, introduced in 2003, applies an area-based daily charge to vehicles entering the central zone, with the objective of reducing private vehicle trips and improving traffic flow in one of Europe’s most congested urban cores [4]–[5]. The London experience highlights several design factors critical to scheme success: clear public communication about pricing rationale, robust enforcement infrastructure, targeted exemptions for certain vehicle categories, and complementary investment in public transport capacity to provide viable alternatives for displaced road users. The scheme also illustrates that congestion pricing is an ongoing policy process rather than a one-time intervention — toll levels, exemptions, and zone boundaries have been revised multiple times in response to changing travel patterns, vehicle technology, and urban policy priorities .

Stockholm is often mentioned since it has tried a congestion tax prior to its permanent introduction. The value of evidence-based evaluation and public response is illustrated in this case. Experience in Stockholm indicates that the degree of public acceptance may be raised if the users see benefits in practice, like travel time reduction and increased reliability. It also demonstrates that the more a scheme’s goals and results are quantifiable, the more acceptable it becomes.

While these examples are helpful, they are not suitable for copying for use in Auckland. Auckland has its own geographical, road network, public transport, political, and equity considerations. So the idea of congestion pricing is generally accepted in the international literature and the need for local modelling and careful system design is evident. This thesis makes a contribution to the local design process by testing a prototype with an Auckland-specific simulation and pricing process.

2.2.3 Relevance to Auckland

Auckland’s congestion is shaped by a distinctive combination of factors: a dispersed, low-density urban form that promotes car dependency, corridor capacity constraints at key bottlenecks, and limited rapid transit coverage across much of the network [1]–[2]. These characteristics make congestion pricing both relevant and challenging to implement. A poorly designed scheme — one that applies charges without adequate alternatives or equity safeguards — risks placing disproportionate financial burdens on lower-income commuters who have few viable substitutes for private vehicle travel [29]–[32]. Conversely, a well-designed system that targets pricing at the corridors and time periods where congestion is most severe, and reinvests revenue into transport alternatives, could meaningfully improve travel time reliability and network efficiency [10]–[12]. The appropriate design for Auckland therefore depends on detailed local modelling, which is one of the motivations for developing the simulation-based prototype presented in this thesis.

In Auckland, a data-driven approach is particularly crucial as the traffic conditions may differ from one corridor to another, and between dawn and dusk [1], [2], [87]–[90]. These variations may not be represented in a single fixed price. The predictive pricing system may be able to aid more focused interventions where and when congestion is likely to occur. But this would have to be transparent, be able to be audited and be under policy protection, which could include tolling caps, exemptions or reinvestment in alternatives to driving.

There are also context reasons for using simulation in the Auckland context. A simulator can be used in the development of a technical workflow before any real system is operated, which does not involve the users of the road. It is possible to test various levels of demand, traffic situations, and pricing reactions in a safe environment with simulation. That is why the present study is based on SUMO-generated data for use as a basis for forecasting and pricing prototype.

2.3 Comparative Analysis of Forecasting Models

The relative performance of forecasting models in traffic applications depends strongly on the characteristics of the input data, including its temporal resolution, degree of aggregation, and the presence of nonlinear patterns [40]–[46]. Statistical models such as ARIMA are well-established in transport forecasting for their interpretability, computational efficiency, and strong performance on

stable, linearly structured time series. However, their ability to capture nonlinear traffic dynamics — arising from incidents, signal interactions, network spillback, and irregular demand patterns — is limited by design. This boundary motivates the comparison with machine learning and deep learning approaches, which are theoretically better suited to nonlinear forecasting tasks but carry greater data and computational requirements [47]–[61].

With the ability of capturing nonlinear relationships and interactions between features, machine learning approaches like XGBoost are gaining in popularity. XGBoost has a special advantage when there are a large number of explanatory variables and the feature engineering is well thought out. However, in small datasets and in the presence of temporal dependency in the prediction problem, the performance of XGBoost can suffer.

The aim of deep learning models is to discover long-term temporal relationships in sequential data and these models in particular are developed to capture these relationships, such as LSTM networks. LSTM models have been shown to have good performance in traffic forecasting problems with large and complex datasets. These models, however, use more complex computational resources, need more training data and more training time than do statistical approaches.

The highly aggregated nature of the corridor-level traffic information has been reported in several studies to be able to make more simplified models, like the ARIMA, perform with the same accuracy as more complex deep learning models. In the same way, it seems natural to select a model based not only on the level of complexity but also on the granularity and nature of the data set.

2.3.1 Statistical Forecasting Models

Despite the fact that statistical forecasting models are losing their significance in traffic research due to their limited applicability, they are still significant in their transparency and efficiency, and they are relatively easy to interpret [40]–[43], [97]. One of the most popular forecasting techniques for time series data is ARIMA. It is a combination of autoregressive terms, differencing and moving-average terms, and models temporal patterns in ordered observations. For traffic forecasting, ARIMA may be applied under conditions of a reasonably stable series and linear dominant patterns in the main traffic characteristics such as travel time, speed or flow in the short term.

One advantage of ARIMA is its simplicity. It needs not be a large number of input features and the structure of the model can be mathematically explained. This is beneficial in transport decision making as agencies frequently require models that can be communicated to technical and non-technical audiences. The models are also fast to train when computational resources are limited, which is useful for ARIMA modeling.

The downside is that ARIMA cannot be easily applied to deal with complex nonlinear relationships. Numerous factors influence traffic systems and are interacting with each other, including incidents,

weather, signal timing, driver behaviour and network spillback. If these interactions are influential, a simple linear model based on the average might not perform as well. However, in data that is aggregated to the corridor level, ARIMA may yet be competitive because it is able to smooth out the noise and local fluctuations. Thereby, ARIMA has been considered an appropriate baseline in this thesis.

2.3.2 Machine Learning Forecasting Models

Machine learning algorithms have been extensively employed in the domain of transport forecasting owing to the fact that they can find relationships between the variables without requiring any assumptions that are necessary for conventional statistical algorithms [60], [61], [98], [99]. The XGBoost algorithm is a perfect example of machine learning algorithms. XGBoost creates a series of decision trees where each decision tree tries to minimize the mistakes made by previous trees. Thus, it becomes possible for the model to capture non-linear relationships between features.

XGBoost can process lagged travel-time data, day-time data, day-type data, speed, density, flow and other engineered features for traffic forecasting. The model is especially helpful where there are more than one explanatory variable. In particular, if the travel time is determined by the previous interval as well as the time of day and the trend of congestion, XGBoost can learn these interactions well.

But XGBoost is very feature engineering driven. Unlike LSTM, it cannot automatically grasp the order of the sequence, but to obtain temporal feature, it must be given manually. The model may not be able to outperform simpler methods if the input data set is small or if it contains only a simple travel-time sequence. This is relevant to the present study as the simulated corridor data set is limited. XGBoost is thus added for comparison as an interesting tool, but its performance is understood with respect to the features provided.

2.3.3 Deep Learning Forecasting Models

The models of deep learning are used widely in intelligent transportation research, due to their ability to be trained on large datasets to learn complex patterns [47]–[59]. LSTM networks are particularly relevant for time-series data because they enable the networks to remember over different time steps. Their memory cells and gating mechanisms enable the model to determine what information it keeps, updates or forgets.

LSTM networks are effective in traffic forecasting scenarios where traffic congestion changes over time, and where past traffic patterns affect the present. An example of a queue that might impact travel time for multiple future intervals could be one that develops on a corridor. A model which could learn the aforementioned temporal dependency may give better predictions than a model

using only the latest value.

The main drawback of LSTM is that it requires a lot of data and computations. But, when the number of observations is limited and the deep learning models are too complex for the data, overfitting can occur. In addition, they need a longer training period and careful parameter tuning. LSTM might be more useful in a real-world deployment, where the sensors continuously provide input. It should be used with caution in a small prototype data set.

Therefore, the current thesis doesn't assume that the most complex model is the best model to use and instead compares ARIMA, XGBoost and LSTM. This is important because the models should be accurate but also practical, explainable, and suitable for the data available for an intelligent transport system.

Table 2.1: Comparison of Forecasting Models

Model	Strengths	Limitations	Computational Complexity
ARIMA	Simple, interpretable, efficient for stable time series	Weak nonlinear modelling capability	Low
XGBoost	Handles nonlinear relationships and feature interactions	Requires feature engineering	Medium
LSTM	Captures long-term temporal dependencies	Computationally intensive and data-hungry	High

Table 2.1: Comparison between the three categories of models considered in this study, including their respective advantages, disadvantages, and computational complexities. ARIMA provides interpretability with lower computational costs; XGBoost provides nonlinear modeling of features with moderate computational costs; LSTM provides long-term temporal learning with higher computational costs. Model choice should depend on the properties of the data rather than on computational cost alone, as explained in Section 2.3.

2.4 Traffic Forecasting Techniques

Proper short-term traffic predictions are a prerequisite for proactive congestion control, since any pricing decisions made from observations rather than predictions are always going to lag behind the actual congestion [44]–[46]. Traffic prediction literature has progressed through three main generations of forecasting approaches. Statistical methods, mainly ARIMA and seasonal versions thereof, prevailed in early studies and remain applicable for aggregate and stable traffic time series [40]–[43]. Machine learning techniques, represented primarily by gradient boosting, such as XGBoost, allowed for non-linear feature interactions and multiple feature inputs in forecasts

[40]–[41]. Deep learning architectures, exemplified by LSTMs and, more recently, graph neural networks and transformer models, took forecasting to the next level by enabling capturing of spatial and long-range temporal dependencies in complex road networks [47]–[59]. Each generation has brought improvements to traffic forecasting capabilities; however, each has also introduced new challenges of data availability and computation demands, which is especially important for prototype implementations that work with limited data.

The deep learning techniques, especially the **Long Short-Term Memory (LSTM) networks** have contributed to the further improvement of the prediction capacity as they can capture long-term temporal dependencies [49], [51], [100], [101]. The LSTM networks are especially good with sequential data and are known for their great accuracy in predicting traffic. However, they require large volumes of data and powerful computational resources. While the above advancements were done, research showed that even in the case of high levels of both traffic data and traffic smoothness, one can use simpler models such as ARIMA just as effectively as complex models.

2.5 Data Aggregation and Corridor-Level Analysis

Typically, traffic data is sampled at a high rate, which can make it difficult to model directly and noisy [62]–[72]. A more usual method of making the series more stable, and easier to interpret, is to aggregate the data into intervals, such as 5, 10 or 15 minutes. In this thesis, 10-minute aggregation is employed since it offers sufficient detail to make short-term predictions as well as alleviates some of the noise which is present in one-minute edge outputs. Using the level analysis of the corridor also allows the problem to be simplified. No individual road elements are individually modelled but some active road edges are grouped together to create one road corridor travel-time series. This offers a broader view of traffic flow, and makes pricing test conditional.

The trade-off here is that by aggregation local detail may be concealed. That means that the more complex models could have less complex variations to learn. Data aggregation also has an impact on the fairness and stability of a pricing system. When the interval is too brief, the toll might be subject to too many changes and might become confusing for users. The system may also be sensitive to random variation if there are very short time constants. A longer interval could cause a lag in the system’s response to congestion. A 10-minute aggregation interval represents an appropriate balance between temporal resolution and signal stability — sufficiently granular to support near-real-time pricing decisions, yet broad enough to attenuate short-term noise in the edge-level simulation outputs.

The corridor level analysis is also applicable to many practical congestion pricing designs. Congestion typically occurs as a route or corridor issue, rather than as a single edge issue. A measure at the corridor level would thus be more easily communicated and more easily linked to a tolling decision.

Corridor aggregation must be carefully designed however, and the road edges chosen should be a true reflection of the road space being priced. The congestion index can lead to false conclusions if links appear that are not relevant.

In the future, in other real world applications, corridor-level analysis could be expanded to include multiple corridors, origin/destination information, and alternate routing. This would enable a pricing system to determine if a toll placed on one corridor would cause some of the traffic to be rerouted to another corridor. These are network-wide effects, which are significant in that congestion pricing should not shift congestion from one road to another, but actually decrease overall congestion.

2.6 Challenges in Dynamic Congestion Pricing

Although there is strong support for the theory behind this type of system and success at implementing dynamic congestion pricing internationally, a number of obstacles remain to its implementation [21]–[32]. Perhaps the most significant of these is the question of public acceptance: studies uniformly find that road users see congestion pricing as yet another cost to them rather than a means of managing demand, especially since the gains of reduced travel time, improved reliability, and better air quality accrue to everyone while the cost is paid directly by the user [25]–[27]. Equity issues further complicate this problem because there is evidence that congestion charging systems unfairly discriminate against poorer commuters, who lack alternative modes of transportation and whose incomes absorb more of the tolling costs [29]–[32]. As Eliasson & Mattsson (2006) observe, the impact of congestion pricing depends greatly on the location of employment opportunities and the availability of public transport services.

Technical integration, equally a very large challenge. As live congestion pricing schemes depend on a reliable communications network to link sensors, forecasts, databases, decision engines there are issues of scale, latency, and security. Furthermore, many of today's pricing systems are based on thresholds or fixed rules rather than learning systems like machine learning. With the growing development of bus systems and the changing urban traffic condition, the need of intelligent learning systems is on the rise. Public trust is also a critical factor. Road users are more likely to accept dynamic tolling schemes if they understand why a charge is being applied, how the toll level is determined, and how revenue from the scheme is reinvested into the transport network. It may be considered as not fair if a dynamic algorithm fluctuates the prices without clearly justification regardless of its technical efficiency. Hence, transparency should be an essential element: models should use comprehensible inputs and explain the relationships, for example, predicted travel time, congestion index and toll cap.

Not only are equity implications important for tolls, but also for congestion pricing generally. The richer will be able to pay the toll, while the poorer will have to change their habits. This may not

result in abolishment of the congestion pricing approach, but policymakers will need to come up with measures to ensure that transport decisions are not taken primarily by the poor. Such measures may consist of exceptions, discounts and reinvesting the revenues to public transport; limitation of toll charges; help for some groups. Another problem is the one of privacy and information security. An efficient congestion pricing approach may involve detection of vehicles, license plate recognition, GPS tracking, cellular phones or payment accounts. All these ways of collecting and storing the information may result in privacy implications if the information will not be secured properly and will be available for others. Any future Auckland approach will have to comply with NZ privacy laws and state what information is going to be collected and how long will it be stored.

2.7 Intelligent Transport Systems and IoT Integration

Intelligent transport systems integrate sensing, communication, data analytics, and decision-support technologies to improve the safety, efficiency, and sustainability of road networks [81]–[86]. Although this thesis operates within a simulated environment, the system architecture is designed to reflect the data flow requirements of a real-world pricing application — in which traffic observations from roadside sensors or transport APIs must be processed by forecasting models and communicated to pricing engines, operator dashboards, and road-user information systems in near-real-time. Understanding these integration requirements is therefore essential for evaluating whether the prototype architecture could be extended toward operational deployment.

In this thesis MQTT is used as an example of how the traffic information could be streamed through the basic IoT layer. This is not a production system, but it shows where we could go in the next phase of the project.

2.7.1 Role of IoT in Transport Systems

The growing relevance of Technologies of IoT in Transport can be explained for the fact that they allow users and systems to collect and share information coming from distributed sources [81]–[86], [89]–[96]. In the case of a road network, sources such as loop detectors, CCTV, GPS, connected vehicles, Bluetooth sensors, public transport feeds, and traffic management platforms could give details about speed, traffic volume, occupancy, travel time, and incidents.

An example where the use of IoT is natural for congestion pricing because prices are determined by real time data. If data is stale or not complete, the determined price will not represent the current situation. Hence, if not the whole system, a solid communication layer has to exist. This communication layer must transfer the data from the point of origin to the forecasting model, from the forecasting model to the price generating module and from there to a dashboard or road-user information system.

It was a wise decision to use MQTT in this prototype because it is a lightweight protocol that uses a publish-subscribe architecture, where the publisher pushes the message onto a particular topic, while the subscriber will receive the messages that have been published by the publisher on the particular topic subscribed. In other words, using this architecture means that the data producer will be separated from the data consumer. Therefore, this makes it easier to extend this architecture.

2.7.2 System Integration Requirements

The type of model used is just one part of an intelligent transport system [81]–[86], [89]–[96]. For a real-world solution, the data collected needs to be stored, analysed and communicated along the system in a cohesive manner. The data from each must fit in one another, else inaccurate results given by the model are likely to cause a failure in the system.

Latency is also a critical requirement. The dynamic pricing should occur within a period meaningful for traffic management. If the toll is set too late, the congestion situation may change already. Therefore, real-time, or almost real-time, systems should take account of the transmission delay of data, time to run the model and time to load the dashboard.

Scalability is another aspect. While a prototype system can process one corridor, a city-wide one may have to handle many corridors, hundreds of thousands of vehicles and many data streams. As such the architecture should be modular. This thesis currently uses MQTT and a dashboard demonstration to demonstrate the basic concept of modularity, noting that a production system would need a more secure architecture, better cybersecurity and higher levels of operations monitoring.

2.8 Demand Response and Behavioural Modelling

Only if there is some demand response to the price signal will congestion pricing function efficiently [22], [23], [31], [25]–[32]. Different people will vary their migration to and from the traffic, including, for example, some will travel at a different time of the day, take a different path, use a different vehicle, want to further trips or shorten them, and not travel at all. The extent of this migration will depend on income, trip purpose, ‘being in a hurry’, availability of substitutes and the combination of tolls and time measures.

The thesis uses an elasticity-based model as it is simple (necessary for a prototype) and assumes a link between price change and travel demand. It does not reflect the complexity of driver behaviour, but provides a workable way to introduce an initial behavioural estimate into the pricing workflow.

2.8.1 Elasticity-Based Response

The concept of elasticity is often used to measure the responsiveness of demand to a change in price [22], [23], [31]. For use in congestion pricing, elasticity measures the percentage change in demand that occurs in response to a percentage change in the cost of travel. Where demand is highly elastic a small toll has the potential to have a substantial effect on traffic levels; where demand is inelastic, a toll of any size may only be able to make a limited impact. This is the case where travellers have limited alternatives or where the trip is unavoidable.

An elasticity-based approach is relevant to a prototype since it offers a relatively straightforward connection between the pricing and the responses in traffic. The model would lend itself to calculating the decreases to be expected in trip-making given a toll, and then this demand reduction could then be coupled to the congestion-feedback calculation. This would allow the pricing component to be more than just a toll-only output [25]–[32].

For the purposes of this prototype, a short-run price elasticity of -0.10 is adopted, consistent with values reported in the urban congestion pricing literature for cities with limited transport alternatives, where short-run demand is relatively inelastic [22]–[23] [22]. This value represents the lower bound of responsiveness and is appropriate for an Auckland context where many peak-period travellers have few immediate alternatives to private vehicle use [1], [87].

But these elasticities are only a simplification. In practice, demand response to congestion pricing varies considerably across traveller groups. Trip purpose plays an important role — commuting, freight, student, emergency, and discretionary trips each carry different levels of time sensitivity and substitutability, producing different effective elasticities. Similarly, individual travellers may respond in qualitatively different ways: some may shift their departure time, others may change route or mode, and some may simply absorb the toll cost without altering their behaviour. Hence, the elasticity value has to be read as an indicative value, not as a sort of behavioural prediction.

2.8.2 Behavioural and Equity Considerations

Behavioral modeling is critical because the effectiveness of congestion pricing relies on alternative options [9], [25]–[32]. With public transport options, walking, cycling, and flexible working choices, the toll could be much easier for the users to adapt to. With poor alternative choices, the toll will only result in increased cost burden without sufficient demand reduction. It is particularly critical in Auckland, where alternative modes of transport differ from suburb to suburb.

Thus, an analysis of equity must be included in future efforts. An efficient pricing scheme needs to look into who pays, who receives the benefits, and what happens to the revenue generated. This way, if the revenue is used to fund transit services or upgrades within the corridor, then the scheme

can create broader social welfare. On the other hand, where the use of the revenues generated cannot be seen clearly, then the public might have less acceptance of such schemes.

The behavioural component in this thesis should therefore be understood as an early modelling layer. It demonstrates that demand response can be linked to dynamic pricing, but it does not replace detailed travel survey data, stated-preference studies or agent-based behavioural models. Future research should build on this by using real Auckland travel behaviour data and by testing different user groups separately.

2.9 Research Gap

The reviewed literature indicates considerable advancement in the areas of traffic forecasting, theory of congestion pricing, and design of intelligent transport systems. Nevertheless, there are a number of gaps that need to be filled and justify the need for this study. In particular, the literature on traffic forecasting tends to measure forecasting models in terms of their accuracy without incorporating forecasting results within an operational pricing or demand management scheme [40]–[61]. This problem was noted by Vlahogianni et al. (2014) who state that traffic forecasting has been developed very quickly independently of the operational system which it should support.

Second, existing congestion pricing implementations frequently rely on static or threshold-based pricing rules that cannot adapt to the continuous variability of real traffic conditions. While dynamic pricing frameworks have been proposed in the literature [10]–[17], relatively few studies demonstrate their integration with predictive machine learning models in a complete end-to-end prototype system. Third, the relationship between data aggregation level and forecasting model effectiveness has received limited systematic attention — a gap that is directly relevant to transport agencies where corridor-level aggregated data may be the most readily available input. Fourth, IoT integration in transport pricing is frequently discussed conceptually but rarely implemented within complete system pipelines [81]–[86]. Finally, there is a notable scarcity of congestion pricing research focused specifically on the Auckland and New Zealand transport context, where geographic, demographic, and policy conditions differ substantially from the European and Asian cities that dominate the existing literature. This thesis addresses each of these gaps through the development of an integrated prototype system..

Third, while deep learning models such as LSTM have shown strong predictive capability, many studies do not critically evaluate whether increased model complexity is justified under aggregated traffic conditions. Limited research has examined the relationship between data granularity and forecasting model effectiveness. [81]–[86]

Finally, there are concepts of IoT implementation that have been described extensively, but rarely implemented in end-to-end transport systems pipeline solutions. The majority of previous literature

has studied individual aspects of the problem, such as prediction, communication, or pricing separately.

Finally, there is limited research specifically focused on the New Zealand and Auckland transport context. Most congestion pricing studies are centred on large international cities such as London, Singapore, and Stockholm, with relatively little attention given to scalable prototype systems tailored to Auckland-network-based simulated traffic conditions.

These gaps can be bridged through this research which proposes a holistic framework for traffic simulation, machine learning based forecast, dynamic congestion pricing, behavioral modeling, and MQTT-enabled IoT communication all together.

2.10 Summary

This chapter has reviewed the key bodies of literature underpinning the present study, spanning congestion pricing theory and international practice, short-term traffic forecasting methods, data aggregation and corridor-level analysis, intelligent transport systems and IoT communication, and behavioural demand response modelling. The review establishes that each of these domains has been studied in depth individually, but that their integration into a single reproducible end-to-end prototype — particularly one grounded in the Auckland transport context — represents an identifiable gap in the existing literature. The following chapters address this gap by developing a unified prototype system that connects simulation-based traffic data generation, multi-model forecasting evaluation, dynamic pricing logic, elasticity-based demand response, and real-time IoT communication within a coherent and extensible pipeline. [1]–[20].

Chapter 3

Dataset and Analysis

3.1 Introduction

This chapter describes the data foundation of the prototype system, covering the simulation environment, data collection procedure, processing pipeline, and evaluation metrics used to assess forecasting model performance. Because real-time Auckland corridor traffic data were not available for this study, a simulation-based approach was adopted using the SUMO platform, and this chapter justifies that choice by explaining what simulation can and cannot represent relative to observed traffic conditions. Each stage of the processing pipeline is described in sufficient detail to allow the study to be reproduced, which is an important criterion for data-driven transport research where model outputs depend as much on data preparation decisions as on algorithmic choices. The chapter concludes by defining the evaluation metrics — MAE and RMSE — and the forecasting setup applied consistently across all three models in Chapter 4. [1]–[20],

This chapter describes the methods used by the raw edge level traffic outputs to be translated into a model input corridor dataset. The raw edge level traffic outputs produced in SUMO were extracted as SUMO XML outputs, manipulated into CSV format, invalid records were filtered out, observations were aggregated into 10 minute bins, supervised learning labels were constructed, and the final dataset was split for model training and testing. Each method used was critical because the accuracy of the forecasting models used requires high quality, well structured input data. [40]–[61],

Furthermore, this chapter describes the criteria used to evaluate the forecasting accuracy. MAE and RMSE are used because they are absolute measures of prediction error, and have been used in many short term traffic forecasting analysis prior studies. In summary, this chapter serves as the methodological bridge between the simulated traffic environment and the congestion pricing model to be implemented later on, by demonstrating the transformation from raw traffic data to prediction, pricing and system analysis evidence.

Another purpose of this chapter is to ensure the research can be reproduced. In a data driven transport research, the resulting model is not only a function of the employed algorithm but also

of data collection, filtering, aggregation, and partition for testing. Without a clear description of such steps, it is hard to assess whether a forecasting output is due to model quality or dataset characteristic. Hence, this chapter explains the data pathway leading to the pricing model (Chapter 4) as follows. [73]–[96]

3.2 Data Source and Simulation Setup

Because real-time Auckland corridor traffic data were not available for this study, a simulation-based approach was adopted using SUMO configured with an Auckland road network derived from OpenStreetMap [73]–[80]. Synthetic traffic demand was generated using a constant random-trip rate to provide a reproducible traffic stream for prototype testing. Unless confirmed directly from the final simulation configuration files, the generated demand should not be interpreted as calibrated peak and off-peak Auckland traffic demand. The 48-hour duration was selected to provide a training dataset with meaningful temporal structure for the forecasting models. Simulation offered a controlled and reproducible alternative to live sensor data: external influences such as incidents, weather events, and sensor faults, which complicate early-stage prototype development, could be excluded, allowing the technical pipeline to be evaluated under stable and repeatable conditions. This data had been imported into SUMO (Simulation of Urban Mobility). The simulation was **run for 48 hours** to provide a time-varying sequence for prototype testing.

Simulation was a valid tool for this thesis as it is necessary to control and repeat a traffic environment, and real traffic data can be influenced by a wide range of external influences, such as weather, incidents, special events or sensor faults. Such influences, while important in real implementations, make it difficult for early prototype development. SUMO provides a controlled environment in which the interdependence of traffic conditions, forecasting output and pricing response can be explored.

This simulation setup is not a whole of city wide transport model but simply an Auckland corridor. This is a key assumption. The dataset is intended to serve in prototype development and technical assessment rather than to say the even the full range of road conditions within Auckland has been simulated. However it provides enough variation in the simulated road network to enable the forecasting and pricing pipeline to be tested without keeping the project out of manageable proportions within a thesis.

The 48-hr simulation was used so that the sequence of observation could be repeated over various traffic conditions. A shorter time would not have yielded sufficient information for model learning. A longer duration would have offered little benefit to the prototype at this stage, potentially resulting in excessive simulation time. The resulting data therefore offered a compromise between realism and scope.

It is recognised that the 48 hours of simulation and single corridor scope results in a set of fewer

observations than would be desirable to be able to build a strong machine learning model comparison. The constraint is appropriate to this prototype investigation, and is explicit as a limitation of the investigation in Section 3.4.7 and Section 5.6. The pipeline architecture is extensible and the next obvious extension of the simulation would be to run a multi-corridor simulation for 28 days as outlined in Section 5.7.



Figure 3.1: SUMO simulation network view

Road network used in the SUMO Auckland traffic simulation, rendered from the OpenStreetMap-derived network file (sim/network/auckland.net.xml). The plot shows 5,482 road edges across the

simulated corridor area, with x and y axes representing SUMO network coordinate units. This network formed the basis for the 48-hour traffic simulation from which all corridor-level travel-time data in this study were generated

The network visual confirms the road graph used to generate the simulated edge-level traffic observations. It is included to make the simulation setup easier to inspect alongside the configuration and data-collection description.

No calibration or validation against observed Auckland Transport traffic counts, measured route travel times, or live sensor records was performed in this study. Therefore, the SUMO outputs are used only as a synthetic Auckland-network-based dataset for evaluating the technical prototype. The quantitative results should not be interpreted as validated estimates of real Auckland traffic conditions.

3.2.1 SUMO Experimental Configuration and Reproducibility

The traffic dataset used in this study was generated using the Simulation of Urban Mobility (SUMO) platform with an Auckland road network derived from OpenStreetMap. The traffic-demand file was generated using SUMO's `randomTrips.py` utility, version `v1_25_0+0000-ac6c5a6752d`.

The route-generation configuration used the network file `sim/network/auckland.net.xml` and produced the route file `sim/routes/auckland.rou.xml`. Random trips were generated for a total simulation period of 172,800 seconds, equivalent to 48 hours. A constant trip-generation period of 2.0 seconds was used, corresponding to one generated trip every two seconds across the simulation horizon. A fringe factor of 1 was applied.

The simulation configuration started at 0 seconds and ended at 172,800 seconds with a simulation step length of 1 second. Edge-level statistics were collected using the `edgeData.add.xml` configuration file and were subsequently processed into 60-second and 10-minute analysis intervals.

It is important to clarify that the generated traffic demand was synthetic and used a constant random-trip generation rate. It was not calibrated to reproduce observed Auckland peak, off-peak, or corridor-specific traffic demand. Therefore, the SUMO scenario should be interpreted as an Auckland-network-based synthetic traffic environment used for proof-of-concept system evaluation rather than as a validated representation of actual Auckland traffic conditions.

The route file contains explicit trip definitions with origin and destination edges and departure times. For example, early trips were scheduled at 0, 2, 4, 6, and subsequent two-second intervals, consistent with the configured `period="2.0"` setting. This provided a reproducible demand stream for testing the downstream forecasting, pricing, behavioural-scenario, MQTT, and dashboard components.

The generated route file contained 86,400 synthetic trips, indexed from trip ID 0 to trip ID 86,399. The final trip departed at simulation time 172,798 seconds. This is consistent with the configured two-second trip-generation period over the 48-hour simulation horizon.

Edge-level statistics were exported using SUMO’s edgeData output configuration with an identifier of edgeStats. The output frequency was set to 60 seconds for the complete simulation period from 0 to 172,800 seconds, and the results were written to `sim/outputs/edge_data.xml`. The parameter `excludeEmpty="false"` was used, meaning that edges without vehicle activity were retained in the raw SUMO edge-output file. Subsequent preprocessing therefore needed to determine how empty or inactive edge records were handled before constructing the aggregated traffic-state indicator.

3.3 Data Collection

In the simulation, SUMO edge-output configuration was used to collect traffic statistics at the edge level [73]–[80], [102]. The following parameters were obtained:

- Edge ID
- Time interval (seconds)
- Mean speed
- Travel time

The data were recorded at **60-second intervals**, and this resulted in high-resolution traffic observations. Raw output was initially saved in the XML format (edge data.xml) and later transformed into CSV format to analyze the results. The raw data had about **1.5 million observations**, which provided detailed information on traffic throughout the simulation period.

Each raw observation represented traffic behaviour for a specific road edge during a specific time interval. This level of detail is useful because it preserves local movement information before aggregation. However, the raw dataset is not directly suitable for the final forecasting task because the pricing model requires a corridor-level travel-time series rather than separate records for every individual edge.

The data collection stage therefore produced a rich intermediate dataset. It contained much more information than the final forecasting dataset, but that detail was necessary to build a reliable corridor measure. The large number of observations also made cleaning important, because small errors or invalid records could accumulate and influence the aggregated travel-time values.

The selected variables were kept intentionally simple. Edge ID was required to identify the road

segment, time interval was required to order the observations, mean speed was useful for checking traffic movement, and travel time was the main variable used for forecasting and pricing. Additional variables such as emissions, density or vehicle counts could be added in future work, but the current thesis focuses on travel time because it is directly understandable and closely related to user experience.

3.4 Data Processing Pipeline

A pipeline of multiple data transformation processes was created in order to turn the original microscopic simulation output data into a time series structure appropriate for modeling and congestion index estimation. The processing pipeline aims to solve the core issue arising due to the mismatch between the resolution of the SUMO edge-level data – where each segment is recorded every 60 seconds in thousands of segments – and the need for a corridor-level travel time indicator to be utilized in the pricing model. Each processing stage serves a specific purpose in terms of improving the data quality and structure: extraction of tabular data out of XML, cleaning of invalid and inconsistent records, temporal aggregation, creation of a single corridor-level series as an indicator, labeling of time series as supervised learning task, and chronological split of the data into training and testing subsets to avoid leakage. SUMO generates fine-grained edge-level records, whereas the forecasting models need a clean chronological time series. The pricing model needs a single congestion indicator interpretable at corridor level as well. Hence, the data needed to go through a number of transformations before being ready for modeling [62]–[72].

The main processing stages were extraction, cleaning, temporal aggregation, corridor aggregation, label generation and dataset splitting. Each stage changed the data for a specific reason. Extraction converted the XML output into tabular form. Cleaning removed invalid records. Temporal aggregation reduced noise from one-minute observations. Corridor aggregation created a representative travel-time series. Label generation converted the time series into a supervised learning problem. Finally, splitting created separate training and testing periods so that model performance could be evaluated on unseen observations. [73]–[80].

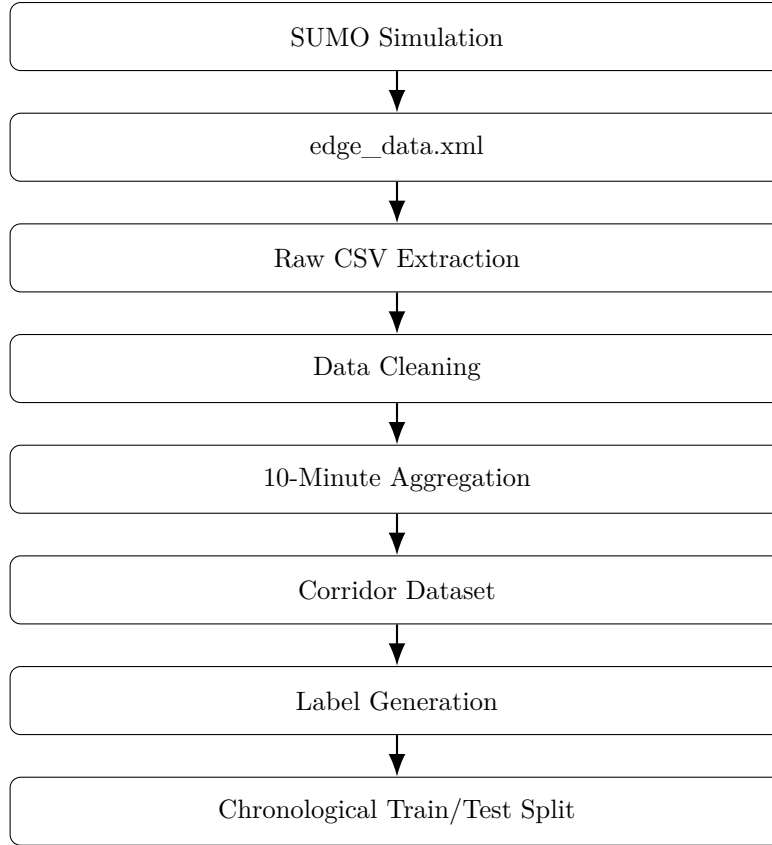


Figure 3.2: Data processing pipeline from SUMO simulation to model-ready datasets

Figure 3.2: 7-stage data processing pipeline converting the raw output of SUMO edges' XML to a supervised machine learning dataset ready for modeling. The stages are presented from top to bottom as follows: SUMO simulation → XML edge output → raw CSV conversion → data cleansing → 10-minute time-series aggregation → creation of the corridor dataset → label creation → chronological splitting into training and testing datasets.

3.4.1 Raw Dataset



Figure 3.3: Transformation of raw traffic data into structured forecasting datasets.

Figure 3.3: Dataset transformation workflow showing three stages of data evolution. Left panel: raw SUMO edge-output at 1-minute resolution with null travel-time values for inactive edges. Centre panel: cleaned and aggregated 10-minute dataset with standardised speed, density, flow, and occupancy fields. Right panel: labelled supervised learning dataset with target travel-time variable (Target TT) representing the corridor indicator 10 minutes ahead. This transformation is the foundation of the forecasting evaluation described in Section 3.5.

The raw data is comprised of edge-level traffic measurements at **60-second intervals** [62]–[80]. This level of detail is helpful, but also noisy and too granular to fit into the pricing model based upon the corridor level.

The raw dataset should be understood as the most detailed data layer in the project. It preserves the original simulation output before any smoothing or corridor-level transformation is applied. Preserving the raw dataset before aggregation is useful because it allows the researcher to inspect the original simulation output and confirm that later results are not produced from a hidden or unexplained transformation. At the same time, raw edge-level data contains several limitations for forecasting. Individual edges can behave differently depending on local geometry, signal timing, turning movements and short-term vehicle interactions. A single edge may also show sudden variation that does not represent the broader corridor condition. If this raw data were used directly, the forecasting task would become more complex and less aligned with the pricing objective [102].

For this reason, the raw dataset was treated as the input to the processing pipeline rather than as the final modelling dataset. The later aggregation steps are not only technical simplifications; they are part of the research design because they convert microscopic traffic observations into a practical

corridor-level indicator.

3.4.2 Data Cleaning

The raw SUMO edge-output XML file was first converted into a tabular CSV structure to enable systematic consistency checking and filtering [62]–[72]. Four categories of invalid record were identified and removed: records with missing edge identifiers, records with missing or non-sequential interval timestamps, records with non-numeric speed or travel-time fields, and records where travel time was zero or negative — values that are physically impossible under normal simulation conditions and likely reflect edges with no active vehicles during that interval. These exclusion criteria were defined prior to inspection of the data to avoid post-hoc filtering decisions that could introduce bias into the cleaned dataset. The time variable was then standardised by converting raw simulation seconds into a consistent chronological index, and duplicate records at the edge-and-interval level were identified and removed. These steps ensured that the cleaned dataset represented a continuous, correctly ordered sequence of valid traffic observations — a prerequisite for all three forecasting models, which depend on temporal ordering for different reasons: ARIMA requires a stationary ordered series, XGBoost requires correctly sequenced lag features, and LSTM requires ordered input sequences for its gating mechanisms to function correctly.

The cleaning stage also standardised the time variable so that each observation could be ordered chronologically. Simulation seconds were converted into a consistent time index, and duplicate records were checked at the edge-and-interval level. Travel-time and speed fields were retained only when they represented valid traffic movement. This was important because the later forecasting task depends on a continuous and correctly ordered time series rather than isolated edge records. Another purpose of cleaning was to reduce noise from very small or inactive edge observations. Some road segments may contain little or no simulated traffic during certain periods, which can produce unstable values. These observations were treated carefully so that the final dataset represented meaningful traffic behaviour rather than artefacts of the simulation output [102].

The cleaning process also helped protect the chronological structure of the dataset. Time-series forecasting requires observations to be ordered correctly, and each target value must correspond to the correct previous values. If timestamps are inconsistent or duplicated, the model may learn an artificial pattern. Therefore, the cleaned dataset was checked to ensure that each interval could be placed in sequence before aggregation. This step is especially important for the later comparison of ARIMA, XGBoost and LSTM. These models handle time information differently, but all of them require reliable input values. A weak cleaning stage would make the model comparison less meaningful because errors could be caused by data problems rather than algorithm performance. The cleaning stage therefore improves both data quality and the validity of the model evaluation.

3.4.3 Treatment of Invalid and Inactive Edge Observations

After the 10-minute SUMO edge-level dataset was created, travel time was calculated only for observations containing valid positive speed and edge-length values. Records with missing values for speed or edge length were removed, as were records where speed or edge length was equal to or below zero.

For each retained edge observation, travel time was calculated as:

$$TT_{i,t} = \frac{L_i}{V_{i,t}}$$

where:

$TT_{i,t}$ is the calculated travel time for edge i at time t , in seconds; L_i is the length of edge i , in metres; $V_{i,t}$ is the corresponding mean speed, in metres per second.

This filtering procedure means that inactive or invalid edge observations were excluded rather than being assigned a travel-time value of zero. This is important because zero-speed observations cannot be used directly in the length-over-speed equation and would otherwise produce undefined or misleading travel-time values.

The resulting dataset retained the timestamp, edge identifier, and calculated edge-level travel time and was then used as the input to the top-20 active-edge traffic-state indicator construction described in Section 3.4.4.

3.4.4 Ten-Minute Edge-Level Aggregation

The raw SUMO edge dataset was recorded at 60-second resolution and was subsequently aggregated into 10-minute intervals for each edge. The raw simulation time, represented in seconds from the beginning of the simulation, was converted into a datetime-like index using the Unix epoch only to support time-series resampling.

Consequently, timestamps appearing as dates beginning on 1 January 1970 in the processed datasets and dashboard do not represent actual calendar dates. They represent elapsed simulation time, where simulation second 0 corresponds to 1970-01-01 00:00:00 for computational convenience. For example, 1970-01-01 00:10:00 represents 600 seconds after the start of the simulation.

For each edge i and 10-minute interval t , the following variables were aggregated using their arithmetic mean:

$$\bar{V}_{i,t} = \frac{1}{n_{i,t}} \sum_{j=1}^{n_{i,t}} V_{i,t,j}$$

where $\bar{V}_{i,t}$ represents the mean value of the corresponding traffic variable within the 10-minute interval and $n_{i,t}$ is the number of available one-minute observations for that edge and interval.

The aggregation procedure calculated the mean of:

edge speed (`speed_mps`); density; flow; and occupancy.

Edge length (`length_m`) was treated as a static road characteristic and therefore retained using the first available value within each edge-time group rather than being averaged.

The resulting dataset contained one record per edge for each available 10-minute interval. Edge-level travel time was not averaged directly from the one-minute travel-time values. Instead, after aggregation, the study calculated travel time using the aggregated 10-minute mean speed and the corresponding edge length:

$$TT_{i,t} = \frac{L_i}{\bar{V}_{i,t}}$$

where L_i is edge length in metres and $\bar{V}_{i,t}$ is the 10-minute mean speed in metres per second. Records containing missing or non-positive speed or length values were subsequently removed before construction of the aggregated traffic-state indicator.

3.4.5 Temporal Aggregation

Once cleaned, the 60 seconds edge level measurements were grouped in **10 minutes time windows** [62]–[72]. Grouping the one minute traffic measures was necessary since the measures could be very inconsistent, especially at the edge level. The 10 minutes grouping helped to strike a balance between the short term variations and random inconsistencies.

For every ten minutes, the travel time values were computed by taking their average on the basis of the pertinent observations. In this way, a smoother series was obtained which was more conducive for making forecasts using ARIMA, XGBoost, and LSTM models. This method also helped reduce the size of the data set and made its use feasible.

The 10-minute interval was selected because the proposed pricing system is intended to respond to near-future traffic conditions rather than long-term daily averages. In a real deployment, this interval could be adjusted depending on data availability, operational policy, and how frequently toll

values are allowed to change. For this prototype, 10-minute aggregation supports the forecasting horizon used throughout the study.

The aggregation procedure also lends itself to realistic tolling behavior. With a pricing mechanism that changes every minute, it could be hard for the drivers to understand or accept. A 10-minute period allows for responsiveness without being too unstable. This also follows the common practice of many short-term forecasts that predict in near-future intervals rather than in whole days.

There is still a trade-off. Aggregation smooths the data and improves stability, but it can remove very short congestion events. This means that the final dataset may not capture every local incident or temporary queue. However, because the thesis focuses on a prototype congestion pricing workflow, the benefit of a stable corridor-level time series is more important than capturing every microscopic fluctuation.

3.4.6 Construction of the Aggregated Traffic-State Indicator

The final forecasting variable was constructed from the processed 10-minute SUMO edge-level travel-time dataset. Rather than defining a continuous physical road corridor, the implementation selected the 20 edges with the greatest number of available observations in the processed dataset. The purpose of this selection was to obtain a stable subset of frequently represented edges for prototype-level forecasting and pricing analysis.

The edge-selection procedure first counted the number of records associated with each edge identifier and ranked the edges by observation frequency. The 20 highest-frequency edges were retained:

$$E_{20} = \text{Top20 (count of observations by edge ID)}$$

For each 10-minute timestamp t , the traffic-state indicator was calculated as the arithmetic mean of the available travel-time observations among these selected edges:

$$I_t = \frac{1}{N_t} \sum_{i \in E_{20,t}} TT_{i,t}$$

where:

I_t is the aggregated traffic-state indicator at time t ; $TT_{i,t}$ is the processed travel-time value for selected edge i at time t ; $E_{20,t}$ is the subset of the selected top-20 edges for which an observation is available at time t ; N_t is the number of contributing selected-edge observations at that timestamp.

A continuous 10-minute time index was then generated between the minimum and maximum

timestamps present in the selected dataset. Where an entire 10-minute bin did not contain an aggregated value, the missing indicator value was filled using linear interpolation. Any remaining missing values at the start or end of the series were subsequently filled using backward filling and forward filling.

The resulting series should therefore be interpreted as a simulation-derived aggregated top-20-edge traffic-state indicator rather than a complete end-to-end corridor journey time. This distinction is important because the selected edges were determined by observation frequency rather than geographical continuity, direction of travel, or route membership. The indicator was used consistently as the forecasting target and as the input to the subsequent congestion-index and pricing-scenario calculations.

3.4.7 Labelled Dataset

To frame the forecasting task as a supervised learning problem, a target variable was generated representing the aggregated corridor travel-time indicator 10 minutes ahead of each observation:

y_{t+10} : travel time 10 minutes ahead

This transformation means that each row of the labelled dataset contains historical travel-time information as the input and the next 10-minute interval’s indicator value as the prediction target. The choice of a 10-minute forecast horizon reflects the intended operational use of the system: a pricing decision should be informed by near-future traffic conditions rather than current observations alone, allowing tolls to be set before congestion becomes severe rather than only in response to it. This labelled structure is applied consistently across all three models — ARIMA uses the ordered series directly, XGBoost uses lagged values as engineered features, and LSTM uses the sequential structure of past observations — ensuring that all models are evaluated on the same forecasting task and that differences in MAE and RMSE reflect model capability rather than task definition. [62]–[80].

The labeled data set is crucial in the process since it transforms the initial time series data into a prediction task. Rather than merely representing the historical information of travel times, the model is trained to predict the future intervals. This is essential for dynamic congestion pricing as the system is supposed to react even before congestion happens.

In practical terms, each observation contains historical travel-time information and a target value representing the next 10-minute travel time. For ARIMA, the ordered time series itself is used to generate the forecast. For XGBoost, lagged values can be represented as features. For LSTM, the sequence structure allows the model to learn temporal dependencies across previous observations. Although the three models use different internal logic, the labelled dataset provides a consistent forecasting task for comparison [102].

Table 3.1: Dataset Summary

Dataset Stage	Rows	Description
Raw 1-minute edge data	1,578,816	SUMO edge statistics
10-minute aggregated data	15,851	Processed edge-level travel time
Corridor dataset	287	Final corridor-level observations
Training dataset	229	Model training
Testing dataset	58	Model evaluation

Table 3.1: Summary of dataset sizes at each stage of the processing pipeline. The raw 1-minute edge dataset (1,578,816 rows) is progressively reduced through aggregation and corridor summarisation to a final forecasting dataset of 287 corridor-level observations, of which 229 are used for training and 58 for testing. The large reduction in row count reflects the aggregation and corridor-selection decisions described in Sections 3.4.2 through 3.4.4

3.4.8 Dataset Splitting

The last data set was divided into a training set and a testing set on a chronological basis:

- **Training set:** 80% (229 observations)
- **Testing set:** 20% (58 observations)

It will avoid any use of future data in the training process and keep the time-series property of the experiment.

Chronological splitting is important in time-series experiments [40]–[46]. If the data is randomized, then past and future samples will be mixed together, and information about future periods will be able to affect the training phase. It will make the evaluation unrealistically positive since the models will be tested on the data that has not been unseen yet temporally speaking. Splitting into 80% of data used for training and 20% for testing reflects the true forecasting process more accurately, since future situations are predicted based on past data.

The training sample was used to train the model, and the testing sample – for the final evaluation of the models. This allows for a reasonable comparison of the models because they are evaluated based on the same unseen time sample. A limited number of data is a limitation in the experiment, but still, it gives enough data for the prototyping of the forecast [102].

3.4.9 Methodological Validity and Data Leakage Prevention

This study’s data is representative of the limitations that are consistent with a first generation prototype study and not representative of a comprehensive operational system evaluation. Here, there are three explicit restrictions that are recognized and dealt with.

First, the final set of forecasts at the corridor level consists of 287 ten-minute forecasts, of which 229 were training and 58 were testing. This is a relatively small number of observations, especially when compared to machine learning models that tend to require more data to effectively generalize across a wide range of traffic scenarios, especially deep learning models like LSTM. This restriction is directly a consequence of the aggregation and corridor-summarisation decisions made in Section 3.4.3 and Section 3.4.4 – the same decisions that yield stable and interpretable corridor indicator needed by the pricing model. This dataset size/indicator stability trade-off is a characteristic of the process of aggregation on a corridor level and not a mistake. It is noted, however, that the data set is not large enough to make statistically definitive comparisons of the models; the results of the forecasts are therefore treated as indicative rather than definitive, and this is reflected in much of Chapter 4. To evaluate the model further [47]–[101] for additional models tested, the data set should be expanded to at least 2,000 observations at the corridor level (about 14 days of simulations could be used to achieve this).

Second, the duration of the simulation (48 hours) has been considered adequate to prove the prototype pipeline in a controlled environment, but is not representative of the amount of temporal variation found in real urban traffic. For a simulation that starts on a Friday and ends on a Sunday, it will take up a 48 hour window of time, with two morning peaks, two evening peaks, and two nights excluding weekend variations. It fails to account for traffic network variation due to weekly seasonality, day-of-week effects, monthly patterns, school terms, public holidays, or special events: All of these are important sources of variation in Auckland’s traffic network [87]–[88]. To start gathering weekly periodicity, 7 days or more would be needed in the minimum, and for seasonal analysis, 28 days or more would be needed. These extensions are both easy to implement in SUMO and considered the top priority in terms of further development of SUMO datasets.

Third, the geographical extent of the simulation is restricted to one representation of the Auckland road network (a corridor). This is suited to a prototype to show end-to-end pricing pipeline, but does not attempt to model multi-corridor interactions, rerouting and network spillover effects as they would occur in a real deployment. The congestion on one corridor can cause the traffic to shift to other corridors in the system, thus generating secondary congestion impacts that one-corridor models would not be able to capture or charge for. Spatial representativeness would be significantly enhanced if the dataset could be extended to include a minimum of 3-5 key Auckland corridors such as the Northern Motorway, Southern Motorway, Northwestern Motorway and arterial alternatives; offering the potential for network-level pricing analysis in future research studies [62]–[72].

These three limitations are coupled: if a longer simulation were run over more corridors, then all three would be solved simultaneously – however, it would be more time intensive. The computational demand of such an extension is not significant; SUMO is able to simulate multiple days of real-world traffic in multiple corridors on standard research computing equipment, and there is no need to modify the structure of the data processing pipeline described in this chapter to support larger inputs. The present data set is, therefore, meant to be regarded as the minimum acceptable base for prototype development and a clear and achievable path is shown to a more comprehensive and stronger upcoming data set.

3.5 Evaluation Metrics

Two standard measures assessed the forecasting models:

3.5.1 Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum |y_{\text{actual}} - y_{\text{predicted}}|$$

MAE measures the average magnitude of the errors [97]–[101].

Mean Absolute Error (MAE) is beneficial because of its ease of interpretation. It represents on average how far the forecasted travel time is away from the actual travel time. As all errors are treated equally in the averaging process once the absolute value is taken, MAE provides a rather simple and straightforward measure of the error to expect. This is important in transport applications as planners are likely to be able to interpret how much forecasting error to expect in the same units as the travel-time variable [40]–[61].

For this thesis, a lower MAE is preferable because it indicates a more accurate prediction of the aggregated edge-based corridor travel-time indicator. Forecast accuracy is directly relevant to the corrected held-out ARIMA-to-pricing evaluation, in which the 58 chronological ARIMA forecasts were passed to the pricing function [62]–[72].

3.5.2 Root Mean Square Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y_{\text{actual}} - y_{\text{predicted}})^2}$$

RMSE emphasizes larger errors and indicates robustness of the model [40]–[61].

RMSE is relevant because it places greater emphasis on larger estimation errors. This is useful for

congestion pricing as any high error could be more damaging as compared to consistently low errors. For instance, if the model underestimates significantly the travel time in a congested period, the pricing system might not react when the demand management is actually required. Similarly, if it overestimates congestion, users may be overcharged [97]–[101].

Using both MAE and RMSE allows a more balanced comparison. MAE shows the typical error, whereas RMSE shows whether the model is susceptible to large errors. A model with a very low MAE and a high RMSE may perform effectively on average, but be a costly failure at various times. Identifying such a problem is critical if one were to employ a forecasting model in a dynamic price optimization system [62]–[72].

3.5.3 Forecasting Setup

The forecasting task in this study is defined as:

- **Forecast horizon:** 10 minutes ahead
- **Input:** historical aggregated edge-based corridor travel-time indicator
- **Output:** predicted aggregated indicator for the next 10-minute interval

Three models were implemented and evaluated:

- **ARIMA (statistical model)**
- **XGBoost (machine learning model)**
- **LSTM (deep learning model)**

The training data set was used to train all models and the testing data set was used to evaluate the models [97]–[101].

The forecasting setup is deliberately scoped to short-term prediction, consistent with the operational requirements of a dynamic pricing system. A 10-minute forecast horizon provides sufficient lead time for a toll value to be calculated, communicated to road users via variable message signs or navigation applications, and acted upon before the predicted congestion conditions materialise. This horizon is also consistent with the aggregation interval of the corridor dataset, ensuring that the forecast target and the input series share the same temporal resolution. Longer forecast horizons — 30 minutes, one hour, or same-day prediction — would be valuable for strategic network management and traveller information applications, but prediction accuracy typically degrades with increasing horizon length as the influence of current traffic conditions diminishes relative to unobservable future events such as incidents and weather. The 10-minute horizon therefore represents an appropriate balance between

operational usefulness and forecasting tractability for the present prototype [40]–[61].

The same forecasting horizon was used for all three models in order to compare them. This helps ensure that the difference in MAE and RMSE can be seen as the difference of the methodology to forecast, not the difference in the forecasting problem which can be the case if the three models are not scheduled to forecast on the same horizon. This setup used in this chapter is then applied in Chapter 4 as a way to approximate the congestion index and the pricing calculation [62]–[72].

3.5.4 Summary

This chapter introduced the dataset generation and processing pipeline, and evaluation approach adopted in this study [1]–[20]. Using high resolution simulation data as input, the pipeline translated raw traffic observations into a structured time-series output for congestion prediction and analysis.

The key point of this chapter is the multiple levels of abstraction that the dataset passes through. The raw edge-level simulation data is cleaned and aggregated into 10-minute time bins before active edges are summarised into an aggregated edge-based corridor travel-time indicator for forecasting. This indicator is critical to the thesis because the subsequent pricing model requires a simpler and consistent traffic-state measure; however, it should not be interpreted as a complete end-to-end Auckland corridor journey time [40]–[61].

In addition, the evaluation logic of the forecasting models has been introduced in this chapter. MAE and RMSE are picked out to take into account respectively on the magnitude and severity of the deviations, and then to set up a good ground for differentiating which model can be applied in practice to the dynamic pricing system. Thesis set up the dataset and evaluation way before the application of the pricing system presents a better link [73]–[96].

This next chapter is the propose methodology, that comprises of the forecasting models, the dynamic pricing algorithm, the demand response modelling and the system integration.

Chapter 4

Proposed Methodology and System Implementation

4.1 Introduction

This chapter presents the proposed methodology and system implementation for the dynamic congestion pricing prototype [10]–[17]. While Chapter 3 explained how the simulation data was prepared, this chapter focuses on how that processed data is used to support prediction, pricing, behavioural response, and system demonstration. The methodology is designed as an end-to-end workflow rather than as a set of isolated modelling tasks.

The proposed system begins with SUMO-based traffic simulation and a processed aggregated edge-based corridor travel-time indicator [73]–[80], [102]. Forecasting models are then used to estimate short-term simulated traffic-state conditions, with ARIMA, XGBoost, and LSTM included to compare statistical, machine learning, and deep learning approaches. In the corrected held-out pricing analysis, the 58 ARIMA forecasts were converted into congestion indices by comparison with a free-flow reference calculated from the 229 training observations only.

After congestion is estimated, a dynamic pricing function converts the congestion index into a toll value. This toll is not intended to represent a final policy recommendation or an operational Auckland toll level; instead, it demonstrates how a technical pricing mechanism could respond to changing simulated traffic conditions. A demand response component is then added to estimate how drivers may react to higher prices, and a congestion feedback step is used to approximate the effect of reduced demand on the simulated congestion indicator [33]–[39].

This chapter further elaborates on the system integration layer, involving the MQTT real-time extension as well as the Gradio dashboard that can be utilized to demonstrate the functionality of the prototype. These elements illustrate how the outcomes of the forecasting and pricing modules can be delivered in practice in an intelligent transport system. The chapter serves as a technical link between the preprocessed data and the resulting output, as it illustrates the role of each element in

the process of congestion pricing.

4.2 Overall System Architecture

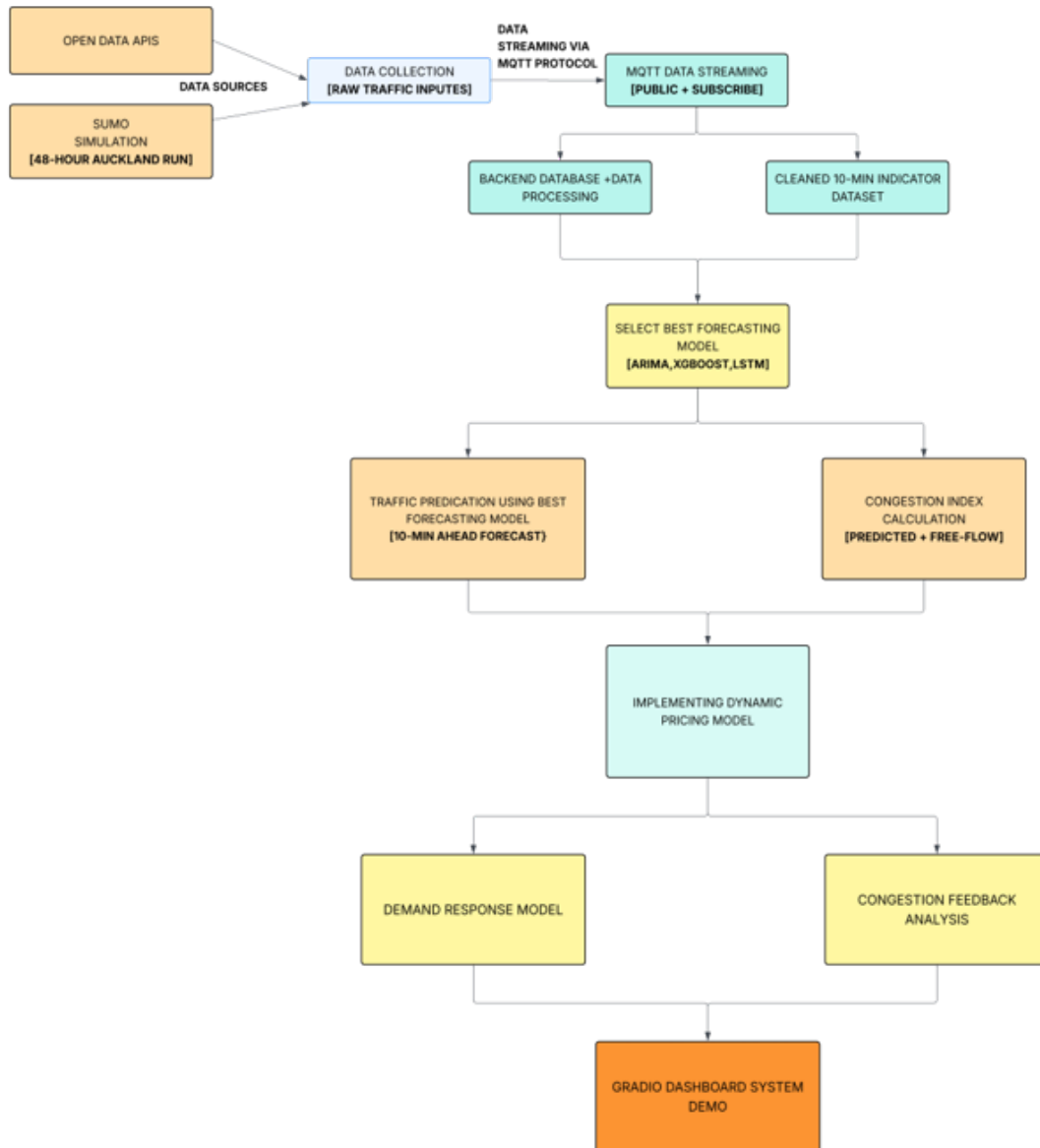
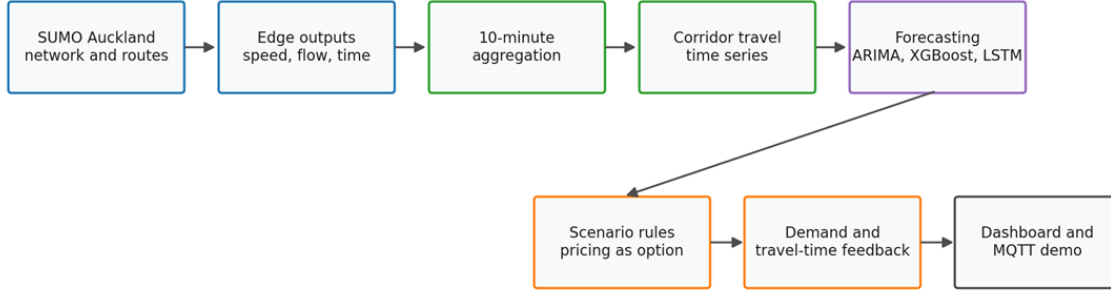


Figure 4.1: System architecture integrating the implementation pipeline and proposal workflow

Figure 4.1: System architecture of the implemented congestion pricing prototype, showing the data flow from SUMO simulation through data processing, forecasting model selection, congestion index calculation, dynamic pricing, demand response modelling, congestion feedback analysis, MQTT communication, and Gradio dashboard demonstration. All components shown in this diagram are

fully implemented and described in Sections 4.2 through 4.5.

Block Diagram: AI-Assisted Congestion Prediction and Decision-Support Workflow



Note: pricing is treated as a configurable scenario layer after prediction, not as the only research objective.

Figure 4.2: Proposed congestion pricing workflow

Figure 4.2: Block diagram of the proposed AI-assisted congestion pricing workflow, showing SUMO Auckland network simulation, edge output collection, 10-minute temporal aggregation, corridor traffic-state construction, forecasting-model comparison, scenario-based pricing, demand and traffic-state feedback estimation, and dashboard and MQTT demonstration. The corrected held-out pricing evaluation connects the ARIMA forecasts for the common 58-observation test interval to the pricing module.

The proposed system follows a structured sequential pipeline: SUMO simulation → data extraction → data processing → forecasting model evaluation → dynamic pricing → demand response → congestion feedback → IoT communication and dashboard demonstration. This linear architecture ensures a clean and auditable trace from raw traffic data to final pricing output, which is important for both research reproducibility and the transparency requirements of any future transport policy application. Each stage receives the output of the previous stage as its input, meaning that improvements or substitutions at any point in the pipeline — for example, replacing SUMO-generated data with live Auckland Transport sensor feeds — can be made without restructuring the downstream components. This modularity is a deliberate design choice that reflects the incremental development path from prototype to operational system. Modularity of architecture. The architecture has been designed as a pipeline in order that each step can be added, tested and optimized separately. In the case of congestion pricing systems for instance, it is interesting to modularize the system as it gathers a variety of tasks: traffic data processing, traffic prediction, toll computation, user/operator response modeling, communications with users/operators, etc. . . Without this module structure the system would be complex with a lack of transparency [1]–[20].

The proposed architecture allows forecasting outputs to feed the congestion index, which then feeds the pricing and demand-response modules. In the corrected evaluation, held-out ARIMA forecasts were passed to the pricing function. The modular structure also allows those forecasts to be replaced with real Auckland sensor data or alternative forecasting models in future work without redesigning the downstream pricing logic [40]–[61].

Similarly the technical approach is separated from the policy interpretation. The forecasting model of the traffic condition does not make the decision on the social acceptability of a toll, and the pricing function transforms congestion into a toll with practical controls in the form of thresholds and caps. This separation is valuable to transport agencies since they require technically sound systems as well as well-understood ones from a governance point of view.

4.2.1 Traffic Simulation and Data Preparation

Traffic data is generated using Simulation of Urban Mobility (SUMO), simulating a 48-hour traffic scenario in Auckland [62]–[80]. Edge-level statistics are collected every 60 seconds and converted into CSV format. The data is then processed into 10-minute aggregated corridor-level travel time, resulting in a structured dataset of 287 observations.

The preparation stage is important because forecasting models are sensitive to the quality and structure of their input data. Raw microscopic simulation outputs contain detailed edge-level information, but this information is too granular for the final pricing objective. A congestion pricing decision is normally made for a corridor, zone, or route rather than for every individual road segment. The data preparation stage therefore transforms detailed traffic observations into a compact travel-time series that can be used consistently by all three forecasting models.

This stage also ensures that the later pricing outputs are based on a single, interpretable unit of analysis. By using the corridor-level traffic-state indicator, the pricing scenario can compare its shifted input proxy with a free-flow baseline and generate a congestion index that is easy to explain. Interpretability is especially important for a prototype intended to support transport decision-making, where system outputs must be explainable to both technical and non-technical audiences, because the system must not only produce numerical results but also show how those results are produced [102].

4.2.2 Justification for Using SUMO

The open-source Simulation of Urban Mobility (SUMO) platform was chosen because of its openness, flexibility, and scalability for research into intelligent transportation systems [62]–[80], [102]. SUMO is a microscopic traffic simulation based upon road maps, capable of simulating realistic vehicles, and their interaction with the road network and traffic.

In comparison of the macroscopic simulation approaches, the microscopic simulation gives more elaborated traffic behaviour information, which is suitable for study on fine details of congestion and travel times variation. SUMO is compatible with OpenStreetMap road network images, which makes it relevant to simulate Auckland urban traffic condition.

One other strength of SUMO is the created reproducible datasets as a result of the controlled traffic conditions. This is useful for evaluating prediction models and reliably testing it on pricing algorithms. The software also allows export of a portion of edge based traffic statistics, which allows integration into higher level data processing and machine learning algorithms.

Using SUMO thus offers one of the best tested and most manageable scenarios for creating and trialling congestion pricing schemes. Another advantage of using SUMO is that it allows experimentation in a safety-net environment, enabling various scenarios to be simulated including traffic flows, data-processing policies, and pricing functions without any danger to real users. This provides a significant advantage for congestion pricing research, since live deployment would have legal, political, economic, and technical risks. A simulation based prototype allows the technical pipeline to be demonstrated before policy-based research is conducted.

Another advantage of using SUMO is that of reproducibility. Once a specific simulation setup is fixed, you can replay the same situation that can then be compared to other modelling decisions. This allows to better check whether improvements in forecasting or pricing accuracy are due to the new approach and not a general evolution of the traffic environment. In academia, reproducibility is even more interesting as other researchers should understand and be able to replicate or modify a given approach.

4.3 Traffic Forecasting Models

Three forecasting models spanning statistical, machine-learning, and deep-learning approaches were implemented for the same 10-minute-ahead forecasting objective. Because lag and sequence construction produced slightly different initial sample counts, the final comparative evaluation was aligned to the common target timestamps shared across all models. ARIMA represents the classical statistical approach and serves as the primary interpretable baseline. XGBoost represents the structured machine learning approach, relying on engineered lag features and ensemble decision trees to capture nonlinear relationships. LSTM represents the deep learning approach, using sequential memory mechanisms to learn temporal dependencies across multiple past observations. The rationale for selecting one model from each category is to enable a methodologically informative comparison: if ARIMA matches or exceeds the performance of XGBoost and LSTM on this dataset, that finding has practical implications for transport agencies where computational simplicity and model interpretability are operational priorities. Conversely, if LSTM or XGBoost outperforms

ARIMA, that would justify the additional infrastructure investment required for machine learning deployment in a live congestion pricing system [97]–[101].

4.3.1 ARIMA Model

The ARIMA (AutoRegressive Integrated Moving Average) approach is one of the forecasting methods based on time-series analysis that models temporal dependencies using autoregression, differencing, and moving average parts [40]–[43]. As a result, the ARIMA model is selected as the key baseline in this research since it is convenient to use for time-series and allows making forecasts based on the aggregated edge-based corridor travel-time measure.

ARIMA was implemented using an ARIMA(1,1,1) specification. The model was trained chronologically without random shuffling to preserve temporal ordering. The original corridor-labelled dataset contained 287 observations, of which the first 80 percentage were used for model development and the remaining observations were reserved for temporal hold-out evaluation. For the final comparative evaluation, predictions were subsequently aligned to the common target timestamps available across ARIMA, XGBoost, and LSTM so that all models were evaluated on an identical test period.

ARIMA(1,1,1) setting is used for the model implemented in the prototype. Previous values of travel time are considered in the autoregressive part, differencing makes the time-series more stable, while the moving average part takes into account the short-term forecasting error. Thus, it becomes possible to recognize patterns from the 10-minute simulated observations of travel time.

The main strength of ARIMA is that it is simple, transparent, and effective for linear temporal patterns. Its limitation is that it does not naturally capture complex non-linear traffic behaviour, sudden incidents, or interactions between multiple explanatory variables. For this reason, it is compared with machine learning and deep learning models in this thesis [97].

Within the prototype, ARIMA provides an important benchmark. If a more complex model cannot clearly improve on ARIMA, then the added computational cost and complexity may not be justified for this dataset. This is a useful question for transport agencies because simple models are often easier to maintain and explain. Therefore, ARIMA is not included only as a basic model, but as a practical reference point for judging whether advanced AI methods provide enough additional value.

4.3.2 XGBoost Model

In the corrected comparative evaluation, XGBoost used the current traffic-state indicator at time t , two lagged indicator values representing $t-10$ and $t-20$ minutes, and the hour and minute of the simulation timestamp. The model was configured with 200 estimators, a maximum tree depth of 4, a learning rate of 0.05, subsampling of 0.8, column subsampling of 0.8, and `random_state=42`. All

XGBoost predictions were evaluated on the same 58 target timestamps used for ARIMA, LSTM, and the naive persistence baseline.

Extreme Gradient Boosting (XGBoost) is a machine learning algorithm that sequentially trains decision trees to create an ensemble model [60]. The final model learns non-linear relationships and interactions between the input features as each new tree tries to fix the mistakes of the previous tree. This research aims to assess whether the XGBoost tree-based model can outperform the traditional statistical model in predicting travel times as short-term forecasting tasks.

In the case of the congestion pricing simulation, the XGBoost algorithm can include lagged features, time-step data, and historical travel-time data as predictors. This is because traffic is affected by past congestion levels and temporal patterns that occur periodically. Also, the learning algorithm is computationally inexpensive and works well with tabular datasets for transportation forecasting experiments [61].

The biggest advantage of using XGBoost is that it can detect non-linear relations and interactions between variables. But it requires careful and thoughtful feature engineering as well as tuning of parameters to perform effectively. Unlike ARIMA, XGBoost does not have inherent temporal structure awareness and requires lag-based feature engineering [98].

In this thesis, XGBoost represents the structured machine learning approach. It is useful for testing whether manually prepared features can capture useful congestion patterns beyond a simple time-series baseline. In a future version of the system, XGBoost could be strengthened by adding weather, incident information, public holiday indicators, roadworks, event schedules, or live speed and volume variables. The current implementation is therefore a first-stage test of tree-based learning within the congestion pricing pipeline [99].

4.3.3 LSTM Model

For the corrected common evaluation, the LSTM used a six-step input window corresponding to 60 minutes of historical traffic-state information and was evaluated on the same 58 target timestamps as the other forecasting methods. The network consisted of one LSTM layer with 32 units followed by a single Dense output layer. It was trained for 30 epochs using a batch size of 16 with the Adam optimiser and mean squared error loss.

LSTM (Long Short Term Memory) is a deep learning technique that has been developed specifically for sequential datasets [49]. LSTM is a variant of recurrent neural networks that use memory cells and gates to determine what information needs to be stored, updated, or erased. Therefore, LSTM is a suitable model for traffic prediction since traffic jams are dependent on past events.

In this study, the LSTM model is applied to the simulated corridor-level travel-time sequence to

learn temporal dependencies from the 10-minute observations. The model is expected to capture more complex patterns than ARIMA, especially where travel time changes gradually across multiple periods or where congestion develops and dissipates over time [51].

The key strength of LSTM is its ability to capture longer-term dependencies in time-series data. Its limitation is that it is more computationally intensive than ARIMA and XGBoost and can require more data to generalise well. Since this thesis uses a relatively small simulated dataset, the LSTM result is interpreted carefully as part of a prototype rather than as evidence of final deployment readiness [100].

The LSTM model is included because a real congestion pricing system would likely receive continuous data streams over long periods. Under those conditions, deep learning models may become more useful because they can learn richer temporal relationships. In the present thesis, the LSTM mainly demonstrates how a sequential deep learning component can be integrated into the same workflow as statistical and tree-based models. This supports the broader aim of comparing different modelling categories within one system design [101].

4.3.4 Naive Persistence Baseline

A naive persistence baseline was included to provide a simple reference for interpreting model performance. For each test timestamp, the baseline predicted the next 10-minute traffic-state indicator using the current observed indicator value:

$$\hat{I}_{t+1} = I_t$$

This baseline represents the assumption that the traffic state remains unchanged over the next 10-minute interval. Including this comparison helps determine whether the forecasting models provide improvement over a simple no-change prediction.

4.3.5 Model Selection Rationale

In this study, the mechanisms or types of models chosen for the forecasting models fall into three categories of prediction method: statistical modelling, machine learning and deep learning [40]–[61]. ARIMA was used for comparison, since ARIMA is a popular and widely-used statistical model for time-series forecasting and is easily interpretable. Thus, the ARIMA model serves as a reference model for deciding if there are techniques that can be considered useful for the aggregated traffic scenarios.

The XGBoost was chosen as it performs very well for structured tabular data problems and can

model nonlinear relationships between the traffic variables. Gradient boosting methods have also been demonstrated to have competitive performances for transport forecasting, without excessive training times [62]–[72].

LSTM is chosen for its capability to capture the long time series dependence between traffic data sequences. LSTM can be used in many fields, especially for intelligent transport problems because it is a time-series learning system which deals with temporal correlation.

To make a comparison, models of various types are employed. This will enable not only the errors of the predictions but also the characteristics of the models, such as their computational complexity and their applicability within a congestion pricing system to be compared. The model chosen for congestion pricing should not just be on the best error score. It must also be dependable, understandable, suitable for operational use and able to cope with the data that is available. An example of this is a more complex model that performs better in one particular test, but isn't easy to use if for instance there was only little data available, and the improvement was significant. This emphasises the need for model selection such that usability is also taken into account [97]–[101].

This comparison addresses a substantive research question within intelligent transportation systems: does increased model complexity always produce better forecasting outcomes? The answer demonstrated in this thesis is no — and that conclusion is itself a product of the AI-driven evaluation framework applied here. The pipeline compares three categories of method drawn from the AI and machine learning literature, evaluates them against consistent metrics on the same dataset, and selects the most appropriate model based on evidence. This process of systematic AI-assisted model selection is a contribution to the field, independent of which individual model performs best. Transport agencies implementing intelligent transport systems benefit from knowing that simpler, more interpretable models may be equally effective under aggregated data conditions, reducing the infrastructure and computational costs of deployment [97]–[101].

4.3.6 Model Evaluation and Results

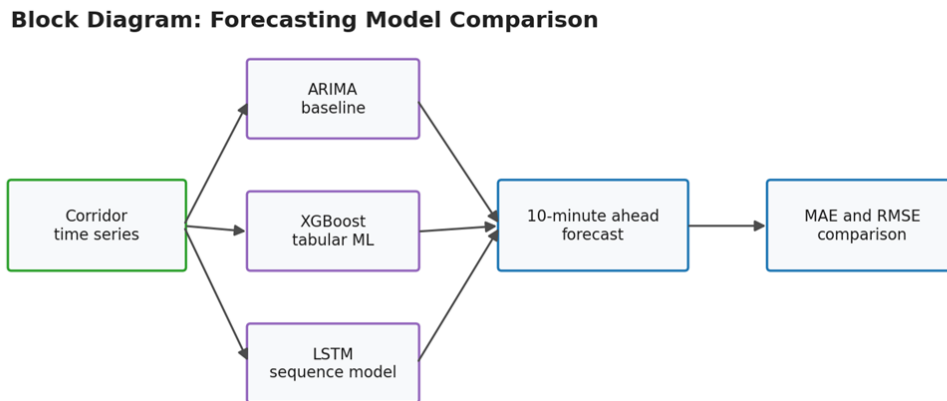


Figure 4.3: Forecasting model comparison workflow

Figure 4.3: Block diagram of the forecasting model comparison workflow, illustrating how the corridor-level travel-time series is passed independently to three model categories — ARIMA as the statistical baseline, XGBoost as the machine learning approach, and LSTM as the deep learning approach — each generating a 10-minute-ahead forecast that is evaluated against the same test set using MAE and RMSE.

For interpretation, naive persistence was included as a simple benchmark, while ARIMA was treated as the main statistical model because it is transparent and appropriate for short-term time-series forecasting. XGBoost and LSTM were compared with these simpler approaches to test whether additional model complexity produced practical improvement. The results should therefore be read as a limited comparison rather than as a claim that all possible forecasting approaches have been exhausted [40]–[46].

Because the final dataset is small and highly smoothed, the evaluation also considers whether model complexity is justified. A complex model that only matches the ARIMA baseline does not necessarily provide a practical advantage, especially when explainability and computational simplicity are important for transport decision-making.

The evaluation focused on the testing set so that model performance could be assessed on observations not used during training. This is important because a model that performs well only on training data may not be reliable for future prediction. In congestion pricing, poor generalisation can lead directly to incorrect toll values. Therefore, the testing results provide the main basis for comparing the forecasting models [62]–[72].

The purpose of evaluation is not only to identify the lowest numerical error but also to understand how each model behaves under the dataset conditions. Since the final corridor dataset is aggregated into 10-minute intervals, the series is smoother than raw one-minute edge-level data. This smoothing can reduce the advantage of more complex models because some non-linear variation is removed during aggregation. The interpretation of results therefore considers both error values and the characteristics of the input data.

Table 4.1: Common 10-minute-ahead forecasting performance on the aligned chronological test set.

Model	MAE (sec)	RMSE (sec)	Test observations
Naive Persistence	0.2453	0.3425	58
ARIMA (1,1,1)	0.1954	0.2508	58
XGBoost	0.2494	0.3235	58
LSTM	0.1962	0.2602	58

Table 4.1 presents the corrected common evaluation of the forecasting models using the same 58 chronological test observations. ARIMA achieved the lowest MAE of 0.1954 seconds and the lowest RMSE of 0.2508 seconds. LSTM produced a very similar MAE of 0.1962 seconds, although its RMSE was slightly higher at 0.2602 seconds. XGBoost recorded an MAE of 0.2494 seconds and RMSE of 0.3235 seconds, while the naive persistence baseline produced an MAE of 0.2453 seconds and RMSE of 0.3425 seconds.

These results indicate that ARIMA provided the strongest overall performance on this particular synthetic aggregated traffic-state dataset. LSTM achieved a comparable MAE but slightly poorer RMSE, while XGBoost did not outperform the simpler persistence baseline in MAE, although it achieved a lower RMSE than the baseline. The results should therefore be interpreted as exploratory findings from one chronological hold-out period rather than evidence of general model superiority.

The forecasting results reveal that both ARIMA and LSTM reach very competitive prediction accuracy overall with aggregated corridor-level data set. The smooth and stable property of the aggregated traffic series diminishes the need for relatively complex nonlinear modeling even in the presence of the more powerful capability of LSTM for sequential learning task.

ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. The aggregation of travel times into 10-minute intervals may have removed some short-term noise and high-frequency variability, simplifying the forecasting problem.

Contraries, XGBoost exhibited a slightly higher prediction error. One hypothesis for this phenomenon could be gradient boosting models’ general tendency to excel in situations with various elaborated

explanatory variables. In our case, the foreseeing problem was to predict future travel times using mainly past travel time observations but lacking abundant multivariate traffic information, which curtailed the model from using nonlinear feature combinations.

Still another important observation from this study is the computational practicality and feasibility. ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. Therefore, model choice in intelligent transportation systems should weigh predictive accuracy, scalability, computational requirements, and interpretability.

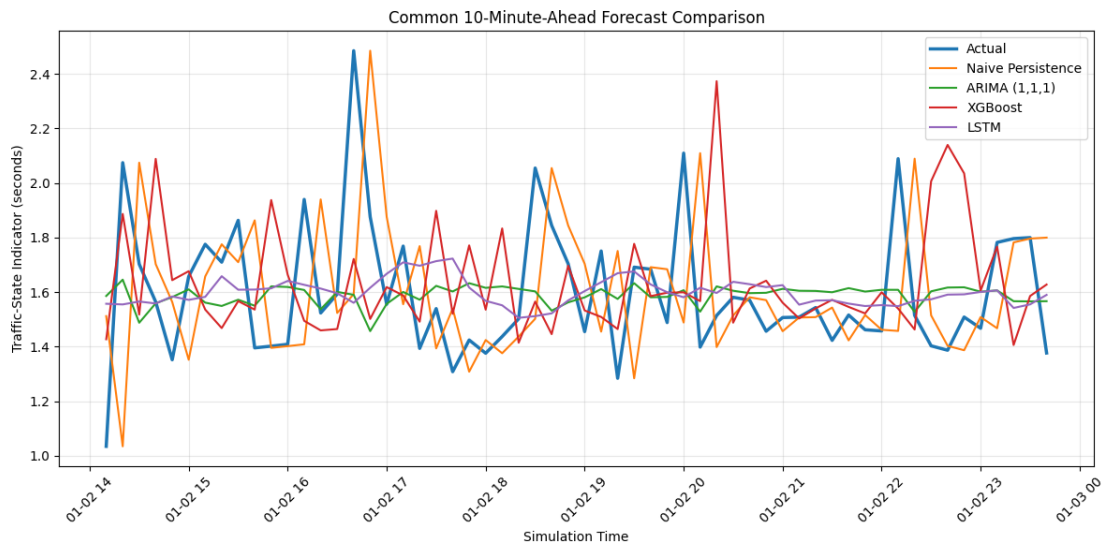


Figure 4.4: Actual and predicted traffic-state indicator values for the common 58-observation test period.

Figure 4.4 compares all forecasting methods using the identical chronological test interval. The common evaluation removes the previous mismatch in test-set size and ensures that MAE and RMSE values are directly comparable across ARIMA, XGBoost, LSTM, and the naive persistence baseline.

Analysis

- ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence.
- XGBoost showed slightly higher error
- The similarity in results is due to corridor-level aggregation, which smooths the data and reduces complexity

4.3.7 Discussion of Forecasting Results

Notably, the forecasting outcomes can be interpreted beyond the numerical values of the error metrics. ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. The outcome is consistent with the importance of selecting models according to the properties of the available data rather than model complexity alone [46], [40]–[97].

Second, the relatively poor performance of the XGBoost model can be explained theoretically as well. Both Chen & Fan (2021) and Zhang & Haghani (2015) prove that the gradient-boosting-based approaches produce better results when there are rich multivariate feature sets with weather variables, incidents, network-wide speed data, among others included in the training procedure [60]–[61]. In the current research, XGBoost was used with a limited number of features consisting mainly of the lagged values of the travel time.

The smoothing effect of 10-minute corridor-level aggregation is a critical contextual factor in interpreting all three results. Aggregation reduces the high-frequency variability that deep learning models such as LSTM are specifically designed to capture [49]–[51]. When this variability is removed, the sequential memory mechanisms of LSTM may provide diminishing returns relative to the autoregressive structure of ARIMA, which is well-suited to stable, linearly dependent time series [40]–[43]. ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. The lower computational requirements of ARIMA remain relevant when considering future operational implementations.

It is important to acknowledge the limitations of these results. The test set comprises only 58 observations, which is insufficient for statistically conclusive model comparison. The reported error values are also very small in absolute terms because the target variable is a highly aggregated simulation-derived indicator with a free-flow baseline of approximately 1.31 seconds, not a real Auckland corridor journey time. These results should therefore be interpreted as confirmation that the forecasting pipeline functions correctly within the prototype, rather than as evidence of the accuracy levels that would be achieved with real traffic data.

4.3.8 Implications for Intelligent Transportation Systems

From our forecasting results, it is to signify that the model performance should be considered based on the characteristic of data in intelligent transportation system [81]–[86]. Though deep learning methodology is widely adopted and advocated as the state-of-the-art approach in this industry, it appears that simple statistical approaches can still perform well with aggregated corridor level data.

This also has practical implications for industrial deployment. Slight model is preferable over

complex models like ARIMA because of less computational cost, ease of comprehension and lesser infrastructural needs. These qualities are desirable for transport agencies looking to have scalable cost-effective forecasting.

Furthermore, the results highlight that aggregated traffic data can heavily distort modeling outcome. As a result, the choice of data collection and aggregation should be regarded as one of the modeling issues which, together with the model selection, are the major factors affecting the forecasting process [89]–[96].

4.4 Dynamic Congestion Pricing Model

The dynamic congestion pricing model applies a transparent rule-based toll schedule to a forecast traffic-state input. Rather than a static charge, the model compares this input with a free-flow reference and increases the toll as the congestion index diverges from baseline conditions. The prototype demonstrates the mechanics of dynamic pricing while enforcing practical policy bounds, including a maximum toll cap [10]–[17].

This model does not aim to provide a precise toll to be directly implemented into a real transport system. Its objective is to illustrate how a traffic-state input may be converted into a manageable pricing signal. In a real transport system, tolls should be based on calibrated congestion levels and assessed for behavioural effectiveness and equity. Consequently, the pricing rule chosen in this thesis is simple, transparent, and flexible.

An earlier pricing-scenario implementation used a one-step shifted traffic-state series as a simplified forecast proxy. The corrected evaluation reported in Table 4.2 instead used the ARIMA forecasts for the common 58-observation held-out period. This correction provides a direct ARIMA-to-pricing scenario while retaining the rule-based and proof-of-concept character of the pricing mechanism.

The pricing module compares each held-out ARIMA forecast with a free-flow reference calculated from the 229 training observations only. Tolls are applied when the index exceeds the specified threshold, increase gradually across the defined bands, and remain bounded by the maximum toll cap. These controls make the pricing mechanism suitable as a transparent decision-support scenario rather than an operational policy model [33]–[39].

In summary, the corrected experiment demonstrates forecasting-model evaluation and direct held-out ARIMA-to-pricing analysis within the same prototype project. The pricing results remain scenario outputs derived from simulated data and should not be interpreted as operational toll recommendations.

4.4.1 Congestion Index and Free-Flow Reference

The corrected pricing mechanism calculated a congestion index (CI) by comparing each held-out ARIMA forecast with a training-only free-flow reference. The free-flow reference TT_{free} was estimated as the 5th percentile of the 229 training observations:

$$TT_{\text{free}} = Q_{0.05}(TT)$$

The congestion index was then calculated as:

$$CI_t = \frac{TT_{\text{ARIMA},t}}{TT_{\text{free}}}$$

where $TT_{\text{ARIMA},t}$ represents the ARIMA forecast for held-out interval t , and $TT_{\text{free}} = 1.3140$ seconds is the training-only 5th-percentile reference value.

4.4.2 Pricing Function

The implemented pricing schedule used congestion-index thresholds of 1.10 and 1.30. No toll was applied when $CI \leq 1.10$. Between $CI = 1.10$ and $CI = 1.30$, the toll increased linearly from NZD 1.00 to NZD 3.00. Above $CI = 1.30$, the toll increased linearly from NZD 3.00 toward a maximum of NZD 6.00, with the maximum reached at $CI = 1.60$ and capped thereafter.

Dynamic Pricing and Behavioural Scenario Parameters

Parameter	Value	Role
Free-flow reference	5th percentile	Reference traffic-state value
Free-flow CI threshold	1.10	Toll activation threshold
Mid CI threshold	1.30	Transition to higher pricing
Severe CI reference	1.60	Point at which toll cap is reached
Minimum positive toll	NZD 1.00	Initial toll
Mid toll	NZD 3.00	Toll at $CI = 1.30$
Maximum toll	NZD 6.00	Pricing cap
Demand elasticity ϵ	-0.10	Assumed short-run response
Baseline generalised cost	NZD 5.00	Price-change denominator
Congestion sensitivity α	0.80	Demand-to-traffic-state feedback
Peak periods	07:00–09:59; 16:00–18:59	Descriptive scenario grouping

Note: The elasticity, baseline cost, congestion-sensitivity coefficient, and pricing thresholds are modelling assumptions used for proof-of-concept scenario analysis and are not parameters calibrated from observed Auckland road-pricing behaviour.

4.4.3 Pricing Results

Table 4.2: Corrected held-out ARIMA-to-pricing scenario results

Metric	Corrected value
Training observations	229
Held-out pricing intervals	58
Training-only free-flow reference	1.3140 sec
Tolled intervals	58
Percentage tolled	100.00%
Average toll when applied	NZD 2.0886
Cumulative toll-index sum	NZD 121.1389
Average peak toll	NZD 2.0708
Average off-peak toll	NZD 2.0966

Table 4.2: Corrected held-out ARIMA-to-pricing scenario results.

The cumulative toll-index sum is the sum of interval-level toll quotations under one-unit-per-interval normalisation and is not actual revenue because charged vehicle counts are not incorporated.

The corrected pricing evaluation was conducted using 58 held-out ARIMA forecasts that were not used for model fitting. The free-flow reference was estimated using only the 229 training observations, producing a value of 1.3140 seconds. All 58 held-out forecast intervals exceeded the configured congestion-index activation threshold and therefore received a positive toll. The average applied toll was NZD 2.0886, while the cumulative toll-index sum across the 58 intervals was NZD 121.1389.

The fact that all held-out intervals were tolled should be interpreted cautiously. It indicates that the selected threshold configuration, together with the synthetic aggregated traffic-state indicator and the training-derived free-flow reference, classified every held-out forecast as above the toll-activation threshold. It does not demonstrate that an operational Auckland pricing system would toll all periods. The result is therefore a property of this proof-of-concept scenario and the selected parameter settings.

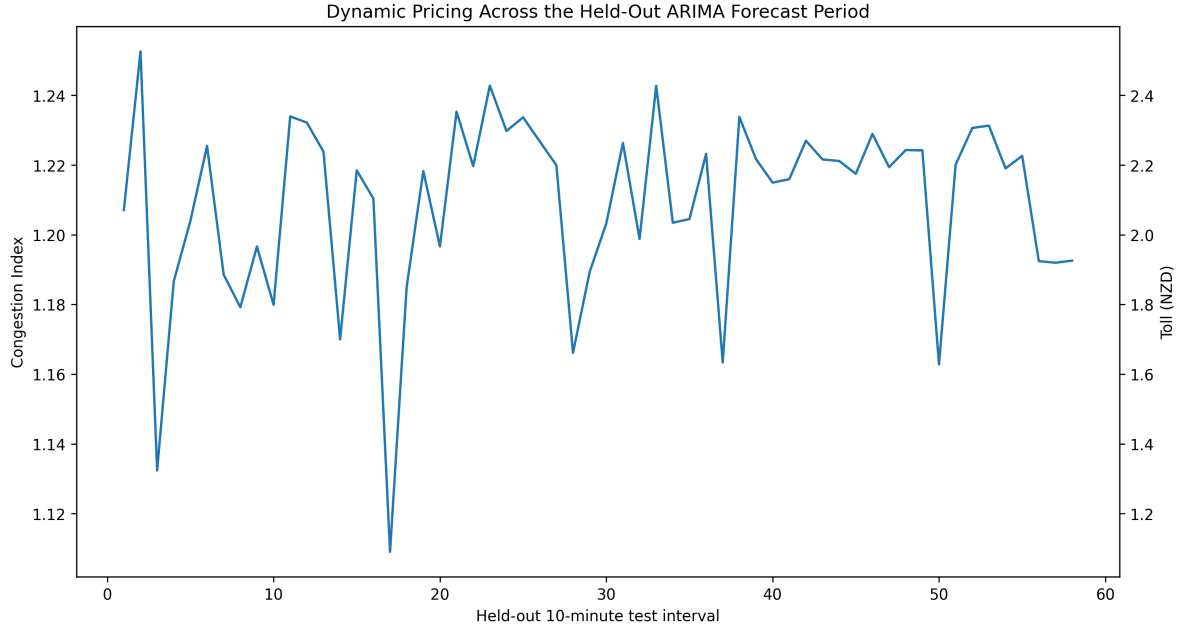


Figure 4.5: Dynamic pricing across the 58 held-out ARIMA forecast intervals.

Figure 4.5: Congestion-index and toll values across the common 58-observation held-out ARIMA forecast period. The figure visualises the corrected ARIMA-to-pricing scenario summarised in Table 4.2. All values are based on simulated SUMO data and should not be interpreted as operational Auckland toll estimates.

Table 4.3: Model-derived demand-response scenario results

Metric	Corrected value
Average scenario-estimated demand reduction	4.1772%
Maximum scenario-estimated demand reduction	5.0518%
Elasticity assumption	−0.10
Baseline generalised cost	NZD 5.00

Table 4.3: Model-derived demand-response scenario results.

4.4.4 Practical Interpretation of Pricing Results

The average toll of NZD 2.0886 when active falls within a range consistent with international congestion pricing implementations. London’s congestion charge operates at a daily flat rate considerably higher than this, while Singapore’s Electronic Road Pricing system applies variable charges that can fall below NZD 2.00 during off-peak periods [3]–[4]. While direct comparison is not

appropriate given that the present study uses a simulation-derived indicator rather than real journey times, the order of magnitude of the prototype toll is not unrealistic for a future Auckland scheme.

For descriptive comparison, peak periods were operationally defined as simulation hours 07:00–09:59 and 16:00–18:59. All remaining intervals were classified as off-peak. These time periods were used for scenario summarisation and should not be interpreted as empirically calibrated Auckland peak-period definitions. Timestamps displayed as dates beginning in 1970 represent elapsed simulation time rather than actual calendar dates.

An important point of the temporal pricing policy is that the mean off-peak toll rate was marginally greater than the peak period toll rate for certain intervals. It shows the occurrence of non-recurrent congestion during the course of the simulation – the period when the corridor indicator exceeds the free-flow level as a result of the simulation process and not a part of the peak hour. This example shows one of the main benefits of dynamic pricing policy compared to static pricing policies: its ability to react to the current situation, and not a preconceived time-of-day [10]–[12].

Incorporating the toll cap in the pricing function was important for policy realism. In the absence of a toll cap, the tolls calculated using the pricing function may be too large during times of serious congestion, which would create equity issues. This consideration was consistent with practice in the literature on managed lane pricing, where the use of a bound on the toll has been commonplace in balancing between revenue collection, demand management, and equity. All the prices generated are just prototypes for illustration purposes and do not constitute operational Auckland toll figures.

4.4.5 Demand Response Modelling

A simplified elasticity-based scenario model was used to estimate potential demand response to the simulated toll. A short-run elasticity parameter of -0.10 and an assumed baseline generalised travel cost of NZD 5.00 were used. For each interval, the relative price change was calculated as:

$$\Delta P_t = \frac{T_t}{C_0}$$

where T_t is the simulated toll and $C_0 = 5.00$ NZD is the assumed baseline generalised cost.

The proportional demand change was then estimated as:

$$\Delta D_t = \epsilon \Delta P_t$$

where $\epsilon = -0.10$. The remaining-demand factor was calculated as:

$$D_t^{\text{remaining}} = 1 + \Delta D_t$$

These values represent scenario-based elasticity calculations rather than observed behavioural responses from Auckland motorists.

Example

For a toll of NZD 1.60:

$$\Delta P = \frac{1.60}{5.00} = 0.32$$

$$\Delta D = -0.10(0.32) = -0.032$$

Therefore:

$$D^{\text{remaining}} = 1 - 0.032 = 0.968$$

Thus, the modelled demand reduction is 3.2%, not 0.03%.

4.4.6 Interpretation of Demand Response

Under the assumed elasticity of -0.10 and baseline generalised cost of NZD 5.00, the corrected held-out pricing scenarios produced an average estimated demand reduction of 4.1772% and a maximum estimated reduction of 5.0518%. These values are deterministic outputs of the assumed elasticity relationship and should not be interpreted as observed behavioural changes among Auckland motorists.

It is also important to contextualise the direction of the demand response. The elasticity model applied here estimates an aggregate reduction in vehicle trips during tolled intervals. In reality, this reduction may represent a combination of trip suppression, mode shift, departure time change, and route diversion — each of which has different implications for network performance and equity [25]–[32]. A more granular behavioural model using stated-preference data or agent-based simulation would be required to disaggregate these responses and assess their individual contributions to congestion reduction. This is identified as a priority direction for future research in Section 5.6.

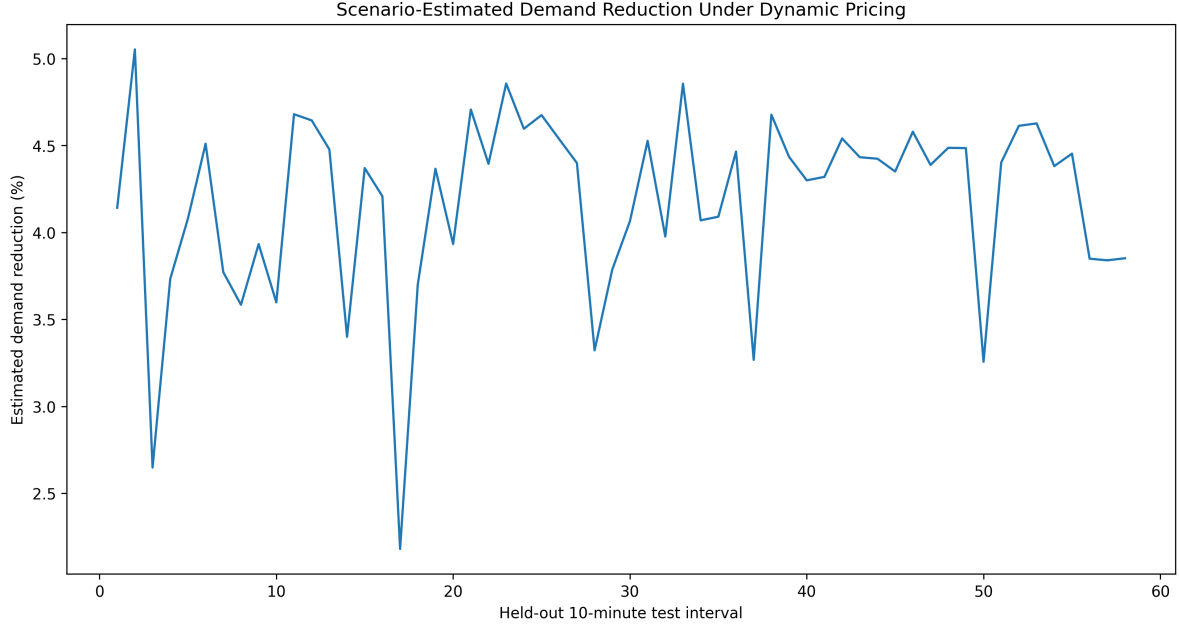


Figure 4.6: Scenario-estimated demand reduction across the 58 held-out pricing intervals.

4.4.7 Congestion Feedback Analysis

A simplified congestion-feedback scenario was applied to translate the estimated reduction in travel demand into a corresponding change in the traffic-state indicator. A congestion-sensitivity coefficient of $\alpha = 0.8$ was assumed. The adjusted traffic-state value was calculated as:

$$TT_t^{\text{adj}} = TT_t [1 - \alpha (-\Delta D_t)]$$

where TT_t represents the original simulation-derived traffic-state indicator, ΔD_t is the proportional demand change produced by the elasticity model, and $\alpha = 0.8$ is the assumed congestion-sensitivity parameter.

Because the demand-change values are negative when tolls reduce estimated demand, $-\Delta D_t$ represents the magnitude of the estimated demand reduction. The feedback calculation therefore reduces the traffic-state indicator proportionally to the estimated demand reduction.

This calculation was implemented as a deterministic scenario model and was not generated by rerunning SUMO with reduced vehicle demand. Consequently, the resulting congestion reductions represent model-derived scenario estimates rather than empirically observed or microsimulation-validated causal effects.

Under this linear assumption, the estimated percentage reduction in the traffic-state indicator is the

congestion-sensitivity coefficient multiplied by the estimated percentage demand reduction:

$$0.8 \times 4.1772\% = 3.3418\%$$

Likewise, the maximum scenario value follows directly from:

$$0.8 \times 5.0518\% = 4.0414\%.$$

The reported 3.3418% and 4.0414% values are therefore mathematically generated from the assumed parameters and are not independent empirical findings.

Table 4.4: Model-derived congestion-feedback scenario results

Metric	Corrected value
Average scenario-estimated traffic-state reduction	3.3418%
Maximum scenario-estimated traffic-state reduction	4.0414%
Congestion-sensitivity coefficient α	0.80

Table 4.4: Model-derived congestion-feedback scenario results.

Applying the congestion-sensitivity coefficient of 0.80 to the estimated demand-response values produced an average model-derived traffic-state reduction of 3.3418% and a maximum reduction of 4.0414%. These values arise algebraically from the assumed feedback relationship and were not validated through a second SUMO simulation with reduced traffic demand. They should therefore be interpreted as scenario estimates rather than measured network effects.

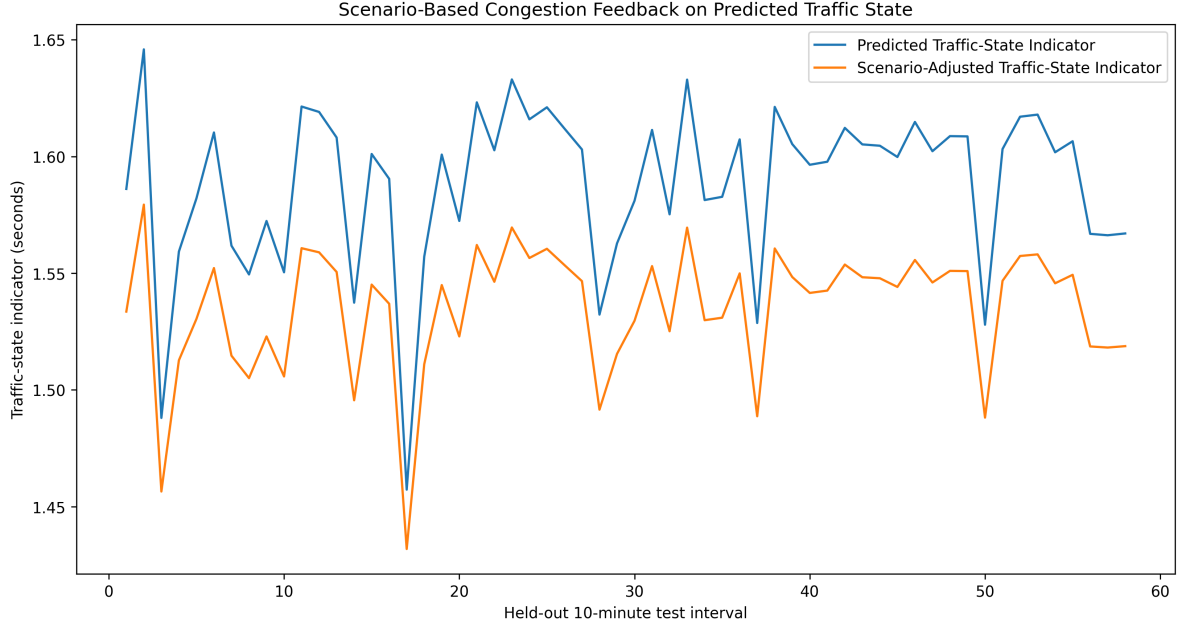


Figure 4.7: Comparison of predicted and scenario-adjusted traffic-state indicators across the 58 held-out test intervals.

Figure 4.7: Predicted and scenario-adjusted traffic-state indicator values across the common 58-observation held-out period. The adjusted series applies the assumed demand-response and congestion-sensitivity relationships. The separation between the series represents a deterministic scenario estimate rather than an empirically observed or microsimulation-validated effect.

4.4.8 Interpretation of Congestion Feedback

The relationship between demand reduction and congestion improvement is non-linear in real traffic networks. Small reductions in vehicle volume can produce disproportionately large improvements near capacity, while reductions during uncongested periods may have little effect [23]. The linear coefficient used here does not represent those network dynamics and should be treated only as a sensitivity assumption.

The feedback component therefore contributes a transparent scenario calculation rather than validated evidence of congestion improvement. Future work should rerun SUMO with modified demand or use observed traffic data to test whether the assumed relationship produces comparable network effects [40]–[61].

4.5 IoT Integration Using MQTT

MQTT was implemented as a local proof-of-concept publish–subscribe communication layer using a Mosquitto broker. The final publisher replayed the 58 held-out ARIMA-based scenario outputs and transmitted JSON messages containing the forecast traffic-state indicator, congestion index, dynamic toll, estimated demand reduction, remaining-demand factor, scenario-adjusted traffic-state indicator, and estimated traffic-state reduction. The subscriber received and displayed these messages through the shared topic `traffic/auckland/corridor`.

This implementation demonstrates end-to-end software-level communication between the forecasting/pricing pipeline and the MQTT messaging layer. It does not represent deployment with physical roadside sensors or a production Auckland transport network.

Communication used the topic `traffic/auckland/corridor`. The publisher and subscriber remained loosely coupled, so neither component required direct knowledge of the other. This supports the modular architecture described in Section 4.2 and allows additional software consumers, such as dashboards or monitoring tools, to subscribe to the same messages [81]–[86], [93]–[94].

The publish subscribe model is particularly beneficial for intelligent transport systems, since most likely future system implementations will consist of a number of data sources and information consumers. Sensors, traffic APIs, simulation modules, pricing engines, dashboards and monitoring systems could all require access to the same information. The publish subscribe architecture helps this due to each message having many subscribers [93].

The publish–subscribe architecture could support future smart-city applications involving traffic APIs, roadside devices, pricing engines, dashboards, and monitoring systems. Such deployment would require security, authentication, redundancy, scalability testing, and integration with operational transport infrastructure; none of these production capabilities were evaluated in the present prototype [94].

4.5.1 System Demonstration Using Gradio

The Gradio interface provides a simple demonstration layer for testing the behaviour of the proposed system [81]–[86]. It allows users to visualise of dataset samples, congestion calculations, and dynamic pricing results.

In terms of academic presentation, the GUI improves transparency of the congestion pricing framework by providing a visual observable link from what the predictions say to what the congestion pricing does.

The use of a lightweight demonstration interface again emphasizes the practical applications of this

research, in addition to theoretical modelling [89]–[96].

A visual interface was developed to present the system working using **Gradio**.

Features:

Main Gradio dashboard for Auckland congestion pricing:

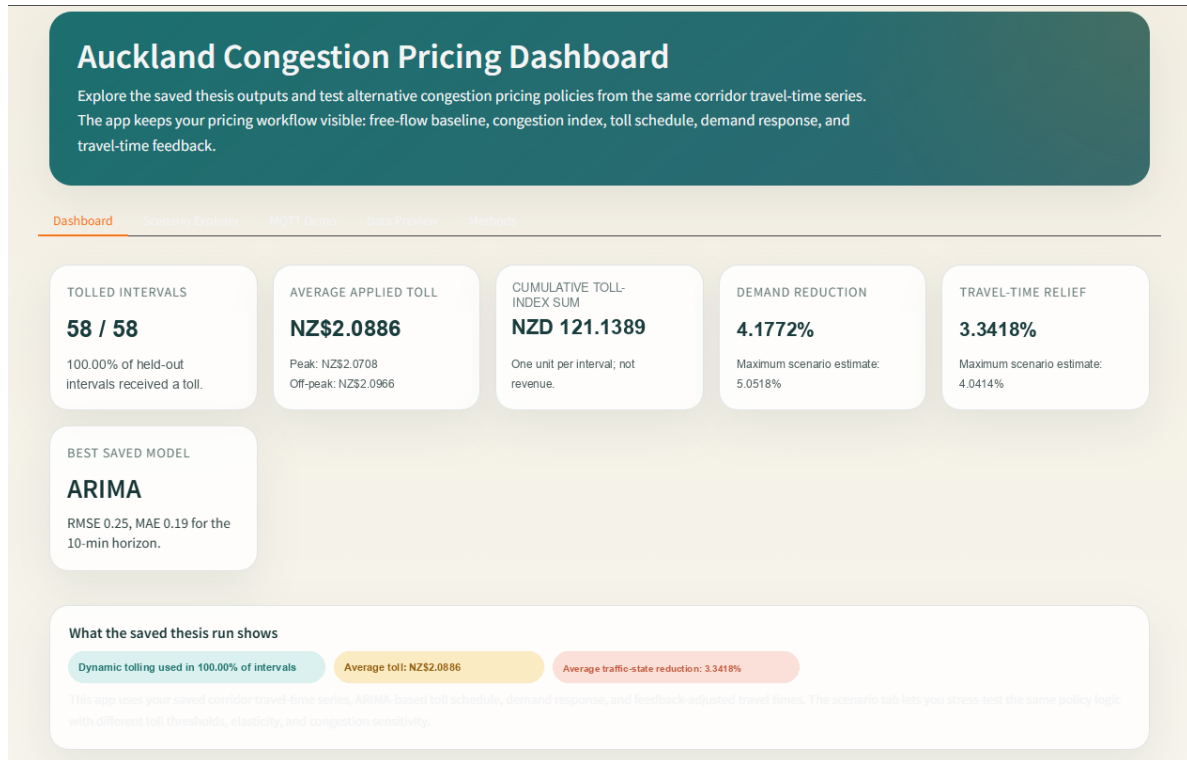


Figure 4.8: Main Gradio dashboard for Auckland congestion pricing

Figure 4.8: Main summary dashboard of the Gradio prototype interface, updated to display the corrected held-out pricing outputs including 58 tolled intervals from 58 held-out observations, an average applied toll of NZD 2.0886, and a cumulative toll-index sum of NZD 121.1389 under one-unit-per-interval normalisation. The cumulative sum is not monetary revenue.

Dataset preview :

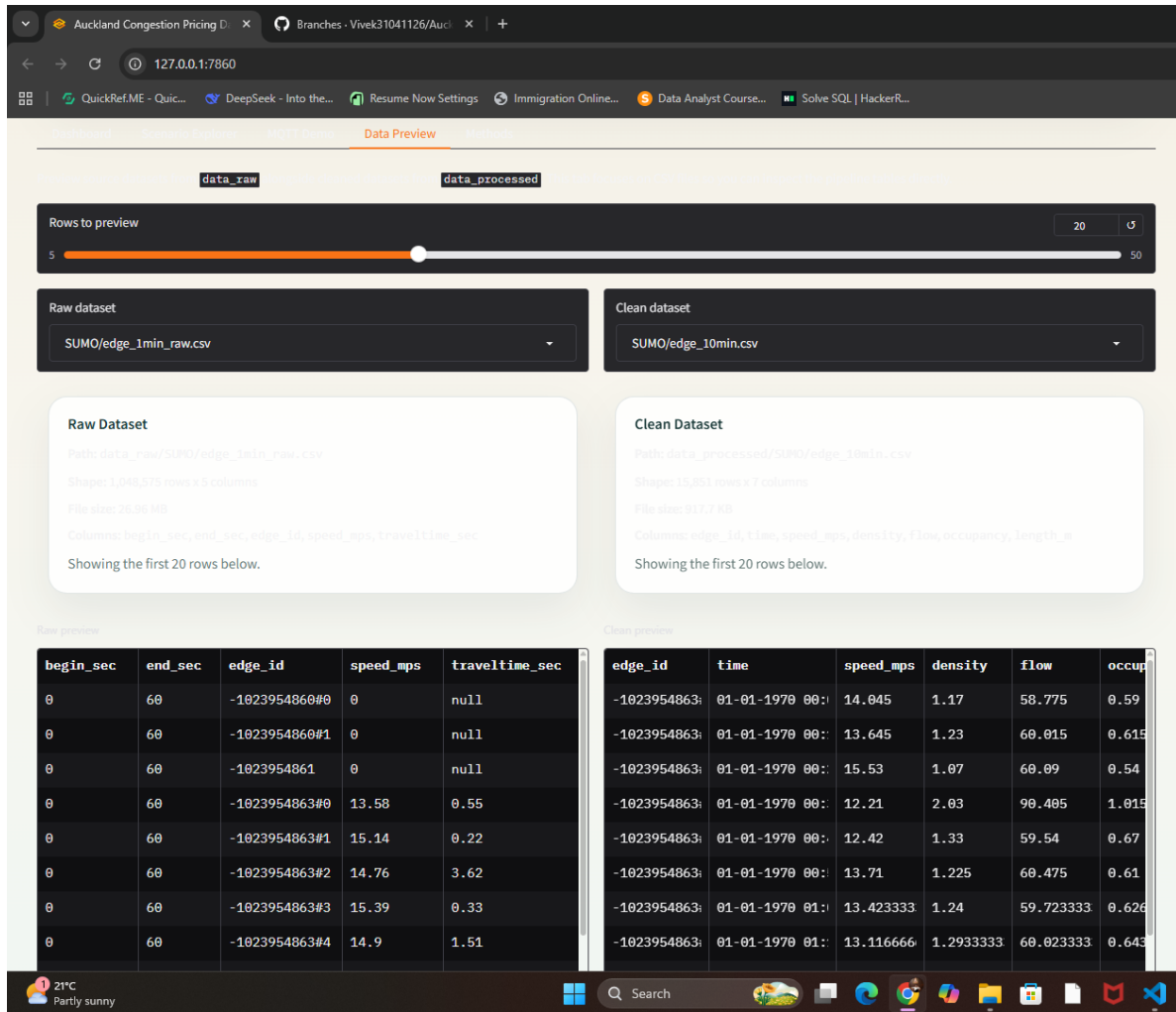


Figure 4.9: Dataset preview dashboard

Figure 4.9: Data preview window of Gradio, showing comparison views of the raw edge-level SUMO dataset on the left (SUMO/edge_1min_raw.csv) and the processed 10 minutes averaged dataset on the right (SUMO/edge_10min.csv). Users can change the number of data rows to be viewed and switch between datasets.

Congestion calculation and Gradio scenario explorer to compare pricing outcomes:

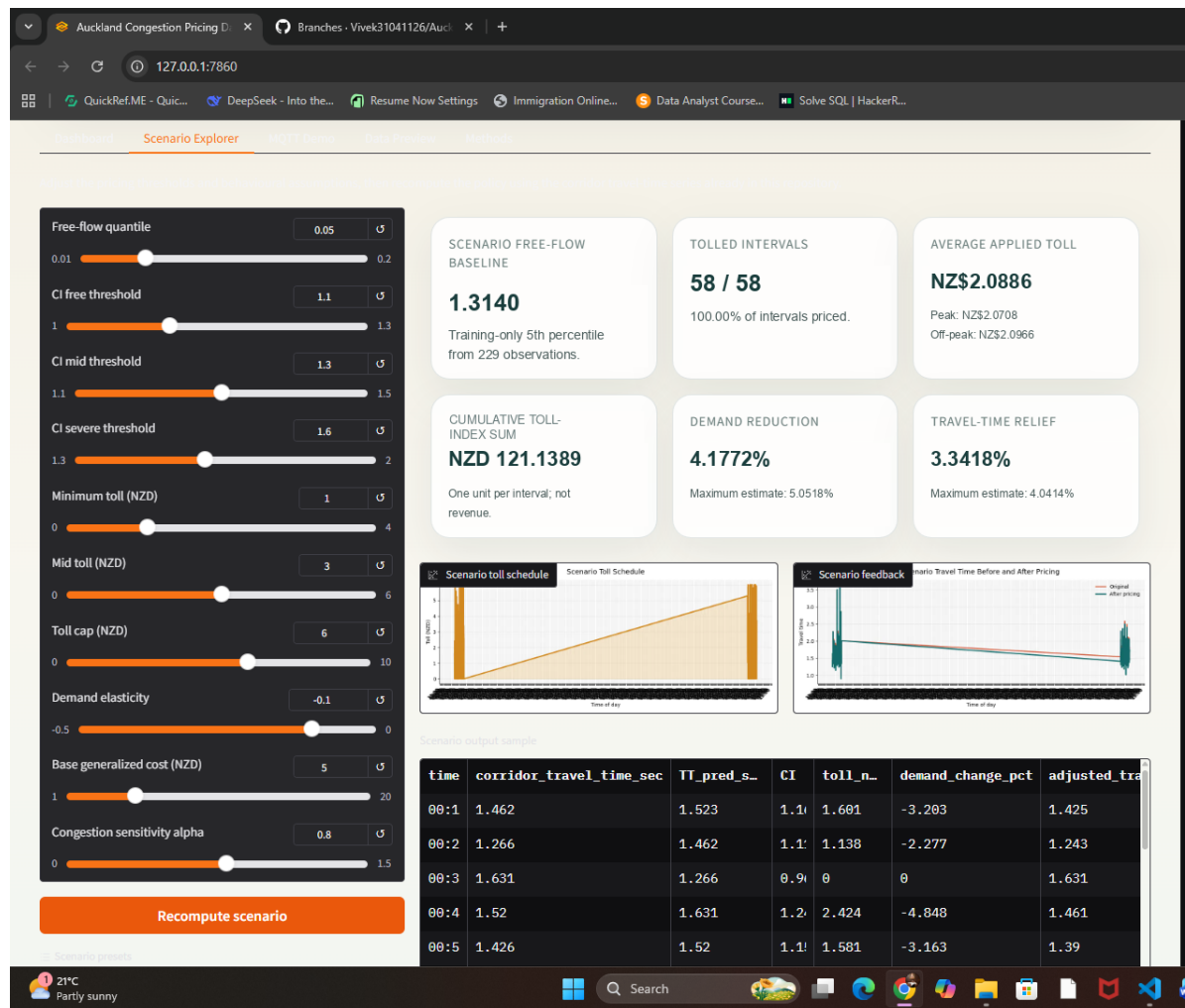


Figure 4.10: Gradio scenario explorer dashboard for comparing pricing outcomes

Figure 4.10: Scenario explorer panel of the Gradio interface, allowing interactive adjustment of pricing parameters including free-flow quantile, congestion index thresholds, minimum toll, mid toll, toll cap, demand elasticity, base generalised cost, and congestion sensitivity. The panel displays updated scenario outputs—tolled intervals, average toll, cumulative toll-index sum, demand reduction, and traffic-state relief—as parameters are adjusted. The cumulative sum assumes one normalized charging unit per interval and is not revenue.

MQTT dashboard interface:

The MQTT demonstration shows software-level JSON message exchange rather than communication with physical roadside sensors or production transport infrastructure.

The dashboard is important because it makes the prototype easier to understand. A congestion pricing system can appear abstract if it is only represented through scripts and CSV files. The interface shows the data, congestion calculation and pricing output in a more accessible form. This supports demonstration, presentation and validation because users can see how changes in traffic conditions influence the toll recommendation.

The dashboard also supports transparency. In future policy applications, a pricing system would need to explain why a charge is being applied. Although the Gradio interface is not a production platform, it demonstrates how an explanatory interface could be built around the pricing algorithm. For the corrected held-out scenario, the interface shows the relationship between ARIMA forecasts, the training-only free-flow reference, congestion indices, and toll levels.

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108 client.on_message = on_message
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110 client.connect(
111     BROKER_HOST,
112     BROKER_PORT,
113     60
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116 print(
117     "Subscriber started..."
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121     "Subscriber started..."
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Figure 4.11: MQTT subscriber receiving final held-out ARIMA-to-pricing scenario outputs through the local Mosquitto broker.

Figure 4.11 provides implementation evidence of the MQTT publish–subscribe communication layer. The local subscriber receives JSON messages transmitted through the `traffic/auckland/corridor` topic. Each message represents one held-out evaluation interval and contains outputs from the final forecasting and pricing pipeline, including the ARIMA-predicted traffic-state indicator, congestion index, dynamic toll, scenario-estimated demand reduction, adjusted traffic-state indicator, and traffic-state reduction. In the final implementation, 58 held-out scenario messages were replayed through the local Mosquitto broker. This demonstrates successful software-level communication

between the forecasting/pricing outputs and the MQTT layer; it does not represent a deployment using physical roadside IoT sensors or live Auckland traffic feeds.

4.5.2 Summary

This chapter has presented the methodology and implementation of the congestion pricing prototype, covering traffic simulation and data preparation, forecasting model development and evaluation, corrected held-out ARIMA-to-pricing logic, demand response modelling, congestion feedback analysis, and IoT system demonstration. The MQTT and Gradio components demonstrate communication and visualisation concepts. The core contribution is therefore a transparent and reproducible collection of connected prototype components, while operational validation with live traffic data remains future work [1]–[20].

This chapter further demonstrates that the real input from the project is actually system integration. It is possible to explore separately each of its individual elements (forecasting, pricing, MQTT communication...), but the benefit of this thesis is to see these elements linked in a single process starting from simulation data and ending with dashboards based system presentation. This gives a much more realistic report than a simple comparison of models: we see how the forecast can assist operation/transport decision [73]–[96].

In parallel, the implementation still is only a prototype. The findings should be taken as technical feasibility evidence of the concept and not as Auckland congestion pricing policy design. More work would be needed to integrate live data, larger simulation scenarios, more advanced behavioural modelling, equity analysis, cybersecurity, institutional policy design and others. These are discussed in more details in the conclusion and future work chapter [40]–[61].

Chapter 5

Conclusion and Future Work

5.1 Introduction

In this chapter, the main conclusions about the research are presented, the contributions made by the proposed system are discussed, the possible limitations are mentioned, and further possible developments are highlighted [1]–[20], [73]–[86]. The goal of the research was to design a data-driven approach to dynamic congestion pricing for Auckland based on simulations, forecasting, pricing, and systems design.

The chapter consolidates the main results obtained during implementation and analysis. It reviews the use of SUMO for generating traffic data, the comparison of ARIMA, XGBoost, LSTM, and naive persistence, and the corrected rule-based pricing scenario generated from the 58 held-out ARIMA forecasts [40]–[61].

Since the system developed in this thesis is a prototype, this chapter also explains the boundaries of the research. The findings are useful for showing technical feasibility, but they should not be interpreted as a final congestion charging policy for Auckland [1]–[2], [87]–[90]. Real-world implementation would require live traffic data, stronger behavioural modelling, public acceptance studies, privacy protection, equity assessment, and policy review by transport authorities.

Overall, the conclusion positions the proposed framework as a foundation for future intelligent transportation research. It shows that simulation, artificial intelligence, pricing logic, and real-time communication tools can be combined into one integrated system, while also identifying the improvements needed before such a system could be tested or deployed in practice.

5.2 Summary of Findings

This research successfully developed and evaluated a complete end-to-end congestion pricing system [10]–[17]. The major findings are summarised below:

1. Traffic Simulation and Data Processing

- A 48 hours traffic simulation is performed using SUMO
- Edge level data collection occurs every 60 seconds
- The data is aggregated into 10 minutes corridor level travel time
- The data set contained 287 observations

2. Forecasting Model Performance

- Three models were applied to predict travel times: ARIMA, XGBoost, and LSTM.

On the corrected common 58-observation test set, ARIMA achieved the lowest overall error with $MAE = 0.1954$ seconds and $RMSE = 0.2508$ seconds. LSTM produced a similar MAE of 0.1962 seconds but a slightly higher RMSE of 0.2602 seconds. XGBoost achieved $MAE = 0.2494$ seconds and $RMSE = 0.3235$ seconds, while the naive persistence baseline achieved $MAE = 0.2453$ seconds and $RMSE = 0.3425$ seconds.

Corridor-level aggregation made the problem easier to predict.

3. Dynamic Pricing Effectiveness

- The congestion index is successfully used to characterize the traffic situation
- Free flow travel time is obtained by the 5th percentile
- Dynamic pricing depends on congestion level
- 58 of 58 held-out pricing intervals were tolled (100.00%)
- Average toll when applied: NZD 2.0886
- Cumulative toll-index sum: NZD 121.1389 under one-unit-per-interval normalisation

4. Demand Response Impact

- Elasticity-based modelling demonstrated measurable behavioural sensitivity to congestion pricing, with demand reductions proportional to the toll level applied in each interval
- Average scenario-estimated demand reduction: 4.1772%
- Maximum scenario-estimated demand reduction: 5.0518%

5. Congestion Reduction

- The deterministic feedback model produced scenario-estimated changes in the traffic-state indicator
- Under the assumed elasticity and congestion-feedback parameters, the scenario model estimated an average 3.3418% reduction in the traffic-state indicator.
- The maximum scenario-estimated reduction in the traffic-state indicator was 4.0414%.

6. System Integration

- MQTT-based communication layer was implemented for prototype data communication
- A Gradio-based interface was developed for system demonstration
- The system demonstrates a prototype communication workflow

5.3 Research Objectives Revisited

This section revisits each of the five specific objectives defined in Section 1.2.2 and evaluates the extent to which each was achieved through the prototype system developed in this thesis.

Objective 1 — Model Development and Evaluation :

For the purpose of training and assessing multiple traffic forecasting models such as ARIMA, XGBoost, and LSTM using traffic simulation data and choosing the best performing model for predicting short-term traffic for congestion pricing. This aim has been entirely met. The three forecasting models and naive persistence baseline were compared on the same 58-observation test period using MAE and RMSE. ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. In the submitted evaluation, XGBoost produced higher MAE and RMSE values than ARIMA and LSTM. ARIMA was identified as the strongest forecasting candidate due to its error values, lower computational cost, and interpretability, and its held-out forecasts were used in the corrected pricing scenario. This is in line with previous studies [40]–[43] which show that simpler statistical models can remain competitive when the input data is aggregated and smooth.

Objective 2 — Data Processing and System Pipeline Design :

To simulate traffic in SUMO for 48 hours, obtain traffic statistics at edge level at 60-second resolution, generate 10-minute corridor-level travel time, and organise data into raw, cleaned, labelled, training, validation and testing datasets. This objective was fully achieved. A 48-hour traffic simulation was conducted using SUMO with an Auckland road network derived from OpenStreetMap. Edge-level traffic statistics were collected at 60-second intervals, producing approximately 1.5 million raw observations. These were cleaned to remove invalid records, aggregated into 10-minute intervals, and summarised into a corridor-level travel-time indicator comprising 287 observations. The final dataset was organised into a training set of 229 observations and a testing set of 58 observations using chronological splitting to prevent data leakage. Each stage of the pipeline is documented in Chapter 3 and illustrated in Figure 3.2.

Objective 3 — Dynamic Pricing Model Development :

To create a congestion index from a forecast traffic-state input compared with a simulated free-flow reference, calculate that reference as the 5th percentile of training indicator values, and create a pricing function implementing congestion-based variable tolls with a toll cap. This objective was achieved through the corrected held-out ARIMA-to-pricing scenario. The free-flow reference was calculated from 229 training observations and was 1.3140 seconds. Across the 58 held-out pricing intervals, all 58 were tolled, with an average applied toll of NZD 2.0886 and a cumulative toll-index sum of NZD 121.1389 under one-unit-per-interval normalisation. These outputs should not be interpreted as operational Auckland toll estimates.

Objective 4 — Demand Response and Feedback Analysis :

To use elasticity assumptions to model driver behaviour, find the effect of dynamic pricing on demand, assess the effects of decreased demand on congestion, and develop measures of travel time and system efficiency benefits. This objective was substantially achieved within the constraints of a prototype study. A short-run price elasticity of demand of -0.10 was applied, consistent with values reported in congestion pricing literature for urban road users with limited alternatives [22]–[31]. The model estimated an average demand reduction of 4.1772 percentage, with a maximum reduction of 5.0518 percentage. Under the assumed congestion-sensitivity coefficient of $\alpha = 0.8$, these values mathematically produce an average scenario-estimated traffic-state indicator reduction of 3.3418 percentage and a maximum of 4.0414 percentage. These are deterministic outputs of the assumptions rather than independent behavioural or congestion findings.

Objective 5 — System Integration and Real-Time Extension :

To create an MQTT-based IoT communication framework, share data using a publish-subscribe pattern, use Gradio to build a user interface, and examine the feasibility of scaling the prototype toward real-world application. This objective was achieved at proof-of-concept level. The final

publisher replayed the 58 held-out scenario outputs through a local Mosquitto broker on the topic `traffic/auckland/corridor`, and the subscriber received and displayed the corresponding JSON messages. Each message included the ARIMA forecast, congestion index, toll value, estimated demand reduction, and scenario-adjusted traffic-state indicator. The Gradio dashboard demonstrated the broader workflow, but the MQTT implementation did not use physical roadside sensors or production transport infrastructure.

5.4 Results

The performance of the proposed travel time forecasting models was evaluated using corridor-level data aggregated at a 10-minute temporal resolution over a continuous 48-hour simulation period [40]–[61], [62]–[72]. Three models were assessed: a classical ARIMA baseline, a machine learning model (XGBoost), and a deep learning model (LSTM). Model accuracy was measured using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

These results must be interpreted carefully. The reported errors are very low because the final modelling target is a highly aggregated corridor-level simulation value, not a raw observed travel time from real Auckland traffic. The dataset is also small, with only 58 test observations. Therefore, the results are useful for checking whether the prototype pipeline works, but they are not sufficient evidence for operational deployment or for claiming that one model would generalise to live traffic conditions.

The ARIMA(1,1,1) model was able to reach a mean absolute error (MAE) of 0.19 seconds and root mean squared error (RMSE) of 0.25 seconds in predicting 10 minutes ahead of travel times. Such performance demonstrates a strong statistical benchmark, showing high levels of temporal regularity of corridor-level travel time data. An XGBoost model, with travel time lagged features and time-of-day features, was able to generate an MAE of 0.25 seconds and RMSE of 0.33 seconds. Even though XGBoost can detect non-linearities, the model is unable to improve over the ARIMA model, which shows limited non-linearities in the data.

ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. The submitted results suggest that additional LSTM complexity did not produce a lower error on this highly aggregated dataset, but no statistical significance test was conducted. The findings should therefore be treated as exploratory evidence that a classical model can remain competitive under low-variance, highly autocorrelated conditions.

The final corrected scenario evaluation used the 58 held-out ARIMA forecasts generated from the common chronological test set. A training-only 5th-percentile free-flow reference of 1.3140 seconds was used to calculate the congestion index. Under the selected pricing thresholds, all 58 test intervals

received a positive toll, with an average applied toll of NZD 2.0886.

The associated elasticity-based scenario produced an average estimated demand reduction of 4.1772% and a maximum of 5.0518%. Applying the assumed congestion-feedback coefficient of 0.80 produced an average estimated reduction in the traffic-state indicator of 3.3418% and a maximum of 4.0414%. These figures are model-derived scenario outputs and do not represent observed or causal reductions in real Auckland traffic.

5.5 Research Contributions

The principal contribution of this thesis is the implementation of an Auckland-oriented proof-of-concept that integrates established methods for traffic simulation, short-term forecasting, configurable pricing, behavioural scenario modelling, MQTT communication, and interactive visualisation. The study does not claim to introduce a new forecasting algorithm or congestion-pricing theory; rather, its contribution lies in demonstrating that these components can be connected within a transparent and reproducible prototype workflow.

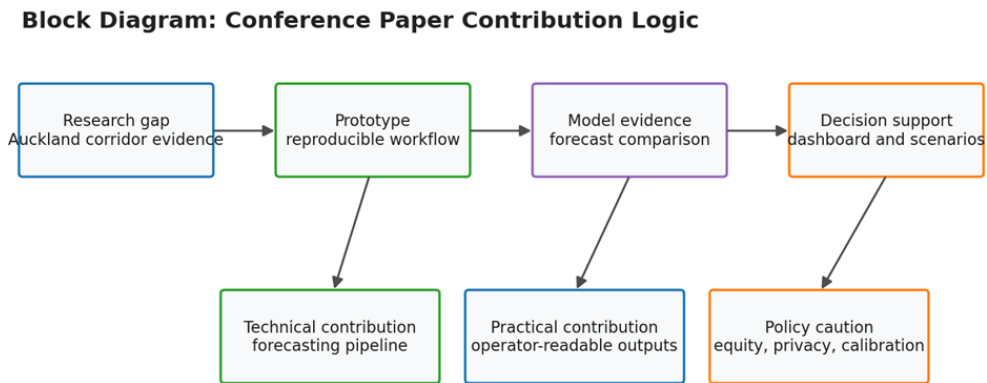


Figure 5.1: Research contribution block diagram

Figure 5.1: Research contribution block diagram showing how the four core outputs of this thesis — Auckland corridor evidence addressing the research gap, a reproducible prototype workflow, model evidence from the forecasting comparison, and decision-support dashboard and scenario tools — combine to produce three categories of contribution: technical contribution to the forecasting pipeline, practical contribution through operator-readable outputs, and a policy caution framework covering equity, privacy, and calibration requirements for real-world deployment.

Development of a prototype AI-assisted congestion pricing project combining traffic simulation, systematic forecasting-model evaluation, corrected held-out ARIMA-to-pricing logic, simplified behavioural modelling, and IoT communication [1]–[20]. The results also show that **basic models like ARIMA can be competitive with more complex models** for this small aggregated dataset. Prototype communication using **MQTT** and a **Gradio demonstration interface** provide building blocks for future smart transport research.

5.6 Limitations of the Study

Despite this, there are still a number of limitations in terms of this approach [1]–[20], [40]–[61], [73]–[96].

The corrected pricing summary connects the 58 held-out ARIMA forecasts to the pricing function. However, this remains a small simulation-based hold-out experiment rather than an operational evaluation. The training-only free-flow reference, pricing thresholds, toll values, and behavioural parameters require calibration and validation with observed Auckland data before practical interpretation.

First, the traffic data used in this study were generated from simulation rather than measured from real-world Auckland traffic sensors. While the SUMO environment is capable of generating such realistic and reproducible traffic data, its traffic behavior may not be truly representative of real-world traffic especially considering higher levels of traffic demand.

Second, the forecasting models were developed using coarse data by aggregating travel time data at corridor level. Aggregation reduces variability and noise however it may hide finer local variations and nonlinear dynamics of the traffic flow, which may affect the accuracy of future predictions.

Third, the forecasting dataset is still relatively small after aggregation, comprising only 287 records, with 229 observations used for training and 58 for testing. This is acceptable for a prototype, but it is weak for strong model comparison. Deep learning models such as LSTM typically require many more records with more diverse features. Therefore, the small dataset in this study limits the reliability of the model comparison and the potential performance gain of complex models.

The forecasting comparison is based on a small simulation-derived dataset and a single chronological hold-out split. A simple naive persistence baseline was included, but no repeated rolling-origin evaluation or statistical significance test was conducted. Consequently, the reported model rankings should be interpreted as exploratory results for this prototype rather than evidence of general forecasting superiority.

These limitations, and perhaps most important from a machine learning point of view, are the combination of the data’s limited size, temporal length, and spatial scope (the corridor). The dataset is small, with just 287 observations at corridor level, from a 48-hour simulation of one Auckland corridor, so the data is not of a large enough size, diverse enough over time and space to draw firm conclusions about the performance of forecasting models or the effectiveness of pricing systems. It includes the fundamental trend of a difference between peak and off-peak traffic and does not include the weekly trend or the effects of day of the week, or the effects of events, school terms or public holidays that cause large fluctuations in traffic in Auckland [1]–[2]. Because of the single-corridor scope, there are no network-wide effects included in the evaluation (rerouting,

spillover or cross-corridor demand substitution). These restrictions do not affect the technical findings and, when interpreting the forecasting accuracy indicators, the price outputs and the estimates of demand response presented in Chapter 4, should be taken into account. All quantitative results should be interpreted as examples of system behaviour and not as predictions for practical use.

Fourth, the travel-time variable is an aggregated edge-based corridor travel-time indicator rather than a full end-to-end Auckland corridor journey time. The simulated free-flow baseline of approximately 1.31 seconds reflects the short edge segments in the SUMO network and should not be interpreted as an actual route travel duration. As a result, the congestion index, toll outputs, demand response and feedback results should be read as proof-of-concept outputs based on a simulation-derived traffic-state indicator. Real deployment would require calibration or replacement using observed route-level travel times from sources such as GPS, Bluetooth, loop detectors, ANPR or Auckland Transport data.

Fifth, the demand-response methodology applied a static elasticity to project the behavioural change associated with congestion pricing. However, a large array of factors might influence driver behaviour, such as income levels, trip urgency, weather, public transportation service levels and route alternatives; therefore, the behavioural model is simplified.

Sixth, the current implementation is not based on a large-scale citywide transport network, but only on a single corridor-level representation. The interactions between different corridors, rerouting behaviour and network spillover were not completely incorporated.

Seventh, although the MQTT integration demonstrates a prototype communication workflow, it represents proof-of-concept real-time communication rather than a production system integrated with physical IoT sensors or transport infrastructure.

Another limitation is that this work focuses mostly on the technical design and evaluation of system performance. Other policy issues related to social equity, public acceptance, environmental aspects and regulatory implementation are not the main focus of this work.

5.7 Future Work

The most immediate direction of future work is the simultaneous expansion of the dataset, along three axes. First, the time frame of the simulation should be increased from 48 hours to at least 28 days (4 weeks), including the beginning of the seasonal pattern emergence and day-of-week effects, and weekend versus weekday variation. For the Auckland network, with 4,032 ten-minute observations in total, a 28-day SUMO simulation would generate around 4,032 ten-minute corridor level observations, fourteen times the current number of observations, which would be enough to

allow statistically meaningful comparison of the model performance using the Diebold-Mariano test for forecast accuracy [40]–[46] and enough sequential data to test whether the architectural advantages of LSTM over ARIMA are more noticeable with more time. Second, the spatial scale should be expanded to include at least three Auckland corridors at once to allow for calculating the congestion index for each corridor, analysing pricing effects at the network level and to provide initial modelling of rerouting behaviour. Third, SUMO should be configured to have different levels of demand for different simulated days, with different peak periods on Mondays, through to the peak on Friday afternoons, to the higher demand on weekends and to the demand during school holidays, to generate a data set that is temporally heterogeneous and not dominated by cycles of the same base demand. The three extensions are technically feasible with the SUMO and Python pipeline as described in Chapter 3, without any new tools or architectural changes, but with the addition of some simulation runtime and minor changes to the demand input files.

There are many possibilities for the development of the suggested framework in the future research [1]–[20]. One important direction involves integrating real-world route-level traffic data from Auckland Transport APIs, roadside sensors, GPS devices, Bluetooth systems, ANPR, or connected vehicle systems. This would enable evaluation of the framework under realistic traffic conditions and allow the aggregated simulation indicator to be replaced or calibrated before any operational policy interpretation.

Further investigation could also be devoted to the development of more sophisticated techniques of prediction like Transformer-based neural networks, TCN, GNN, and reinforcement learning models. Such methods may enhance the predictive ability in highly dynamical urban traffic conditions [73]–[96]. Further important extensions are possible in terms of developing the model from its use at corridor level to that of entire cities’ traffic networks, thereby enabling to study re-routing behavior, network spill-over effects, and multiple congestion regions interaction. Moreover, behavioral modeling can be further advanced through the use of methods like agent-based simulation and/or data-driven estimation of elasticities.

Finally, with regards to system integration, it might be possible in the future to implement an MQTT communication system in real-time, using IoT traffic sensors and cloud-based databases. Additional research may also examine policy-related factors including pricing fairness, socioeconomic impacts, environmental sustainability, and public acceptance of congestion pricing mechanisms within the New Zealand context [40]–[61].

Finally, future smart city implementation can include connected autonomous cars, edge AI, digital twin transport environment, and multimodal transport optimization in terms of buses, rail transit and active transport network.

5.8 Final Conclusion

Auckland’s road network faces persistent and growing congestion that imposes measurable costs on commuters, freight operators, and the broader economy. Conventional demand management tools, including fixed tolls and signal retiming, are limited in their ability to respond to the dynamic and variable nature of urban traffic. This thesis addressed that gap by developing, implementing, and evaluating a prototype data-driven dynamic congestion pricing system that integrates traffic simulation, short-term forecasting, adaptive pricing logic, behavioural response modelling, and real-time communication within a single end-to-end pipeline.

The central research aim was not to design a deployable Auckland congestion pricing policy, but to determine whether a technically sound and reproducible data-driven workflow could support adaptive congestion management. The findings confirm that this is achievable at prototype level. A 48-hour SUMO simulation of an Auckland road corridor provided the traffic data foundation. Three forecasting models — ARIMA, XGBoost, and LSTM — were trained and evaluated on aggregated corridor-level travel-time data. ARIMA and LSTM produced similar MAE values on the common test set, but their RMSE values differed and the results do not establish statistical equivalence. In the submitted evaluation, XGBoost produced higher MAE and RMSE values than ARIMA and LSTM. These results, while based on a small simulated dataset, demonstrate that the forecasting component of the pipeline functions correctly and that the choice of model should be guided by data characteristics rather than model complexity alone. The five research questions posed in Section 1.3 can now be addressed directly. First, the evaluated models can forecast short-term simulated traffic conditions, although the small, aggregated dataset limits general conclusions about model superiority. Second, the corrected pricing scenario demonstrates how congestion indices derived from 58 held-out ARIMA forecasts can be mapped to a piecewise toll function with defined thresholds and a cap. Third, the scenario estimated an average demand reduction of 4.1772 percentage; multiplying this value by the assumed congestion-sensitivity coefficient of 0.8 yields the reported 3.3418 percentage traffic-state indicator reduction. The maximum demand and traffic-state reductions were 5.0518 percentage and 4.0414 percentage, respectively. These values are assumption-derived rather than independent empirical findings. Fourth, simulation, forecasting evaluation, pricing logic, MQTT communication, and dashboard visualisation can coexist within a modular prototype. Fifth, real-world implementation would require live data integration, calibrated route-level travel times, repeated forecasting evaluation, equity analysis, and policy review.

A key finding of this thesis is the relationship between data aggregation and model selection. When corridor-level travel-time data is aggregated into 10-minute intervals, the temporal variation is substantially smoothed, reducing the advantage that complex deep learning models typically hold over simpler statistical methods. This has a practical implication for transport agencies considering intelligent transport systems: the appropriate level of model complexity is determined by the granularity and structure of available data, not by the assumption that more sophisticated models

always produce better results. For Auckland, where real-time sensor data may initially be limited or patchy, this finding suggests that a well-designed ARIMA-based forecasting component could serve as a reliable and interpretable starting point for a live congestion pricing system.

The demand response and congestion feedback components of the prototype, while simplified, demonstrate an important principle: that congestion pricing operates as a closed-loop system in which the price signal, driver behaviour, and network conditions are interdependent. The corrected scenario produced model-derived demand reductions of approximately 4–5 percentage and corresponding deterministic changes in the simulated congestion indicator. These values are based on assumed parameters rather than observed Auckland behaviour. The prototype has clear limitations. The traffic data is simulation-derived rather than observed, the forecasting dataset is small, the travel-time variable represents aggregated SUMO edge outputs rather than full Auckland corridor journey times, and the behavioural model relies on a single constant elasticity value. These limitations are acknowledged throughout the thesis and do not undermine the core contribution, which is the demonstration that all components of a dynamic congestion pricing system — simulation, forecasting, pricing, demand response, and real-time communication — can be integrated into a coherent, reproducible, and extensible prototype workflow.

Future research should prioritise integration with real Auckland traffic data from Auckland Transport APIs, GPS systems, Bluetooth sensors, or ANPR infrastructure. Stronger behavioural models using stated-preference survey data, agent-based simulation, or income-segmented elasticity estimates would improve the realism of the demand response component. Network-wide analysis incorporating multiple corridors and rerouting behaviour would extend the spatial scope of the framework. An equity assessment examining the distributional impacts of dynamic pricing across different income groups and travel patterns in the Auckland context would address one of the most important policy dimensions not covered in this prototype study.

In summary, this thesis demonstrates that a data-driven dynamic congestion pricing system for Auckland is technically feasible as a prototype, that short-term traffic forecasting using simulation data can support adaptive pricing logic, and that the full pipeline from data generation to system demonstration can be implemented in a modular and reproducible way. The work provides a practical foundation for future research and, ultimately, for informed policy development in Auckland’s congestion management strategy.

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